POWER ELECTRONICS (PE) — MODULE 1

The Definitive Master Study Guide, Device Physics Atlas & Derivation Handbook

Course: B.Tech S5 Electrical & Electronics Engineering (EEE) | College of Engineering Trivandrum (CET)

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Foundations of Power Electronics & Thermal Management

Scope, Motivation, and System Architecture

Power Electronics is the branch of electrical and electronics engineering that deals with the efficient conversion, control, and conditioning of electric power using solid-state semiconductor switching devices. In contrast to signal electronics, where information fidelity and linear amplification are paramount, the defining objective of power electronics is **maximum energetic efficiency** ($\eta \to 100\%$), **minimal physical footprint** (high power density), **uncompromising reliability**, and **low system cost**.

It is widely projected that about 80% of all electrical energy will pass through at least one power electronic converter between the generation bus and the end-use terminal (a figure commonly quoted for around 2030; today's share is lower but rising quickly).

Applications Classification of Power Electronics

Power electronic converters form the technological foundation across the following application domains (frequently asked exam question):

1. **Commercial:** Uninterruptible Power Supplies (UPS), elevator drives, HVAC systems, lighting control.
2. **Domestic:** Microwave ovens, washing machines, air conditioner compressors, induction cooktops, LED driver circuits.
3. **Industrial:** Variable Frequency Drives (VFDs) for pumps, blowers, conveyors; DC motor speed control for steel rolling mills, paper mills; arc welding power supplies; electroplating rectifiers.
4. **Telecom:** Switch-Mode Power Supplies (SMPS) for base stations, high-efficiency DC-DC converters for telecom rectifiers.
5. **Aerospace:** Aircraft/spacecraft power management systems, avionics power supplies, electric actuation systems.
6. **Transportation:** Electric/hybrid vehicle traction inverters ($100-300$ kW), railway traction systems, metro/maglev propulsion.
7. **Utility / Power Systems:** High-Voltage DC (HVDC) transmission, Flexible AC Transmission Systems (FACTS), Static VAR Compensators (SVC), STATCOM, renewable energy grid integration (solar PV inverters, wind turbine converters).

Disadvantages of Power Electronics

While power electronics enables high-efficiency power conversion, certain inherent limitations exist:

- **Harmonic Generation:** High-frequency switching generates significant current and voltage harmonics that distort AC mains waveforms, requiring passive (LC) or active harmonic filters.
- **Low Input Power Factor in AC-DC and AC-AC Converters:** Uncontrolled rectifiers draw highly distorted, non-sinusoidal currents with poor displacement power factor, requiring Power Factor Correction (PFC) circuits.
- **Electromagnetic Interference (EMI):** High dv/dt and di/dt switching transients radiate conducted and radiated EMI, necessitating EMI filters, shielding, and compliance with international standards (EN 55022, CISPR 11, FCC Part 15).
- **Low Overload Capacity:** Power semiconductor devices have strict thermal limits and cannot tolerate sustained overload currents (unlike transformers or rotating machines with thermal inertia). Fast-acting protection (electronic fuses, crowbars) is mandatory.

- **Complex Power Regeneration / Bidirectional Power Flow:** Implementing bidirectional energy flow (e.g., regenerative braking in EVs) requires additional anti-parallel switches and control complexity.

A generalized power electronic processing system comprises four interactive subsystems:

1. **Power Circuit (Plant):** Composed of semiconductor power switches and reactive energy storage elements (inductors, capacitors, transformers).
2. **Control Circuit:** Digital signal processors (DSPs), microcontrollers, or analog PWM controllers executing closed-loop feedback algorithms.
3. **Gate Drive Units:** Interface circuits providing galvanic isolation, high instantaneous peak currents, and level shifting to drive the semiconductor gates.
4. **Thermal & Protection Network:** Heat sinks, snubber networks, fast-acting semiconductor fuses, and crowbar circuits to prevent thermal and electrical destruction.

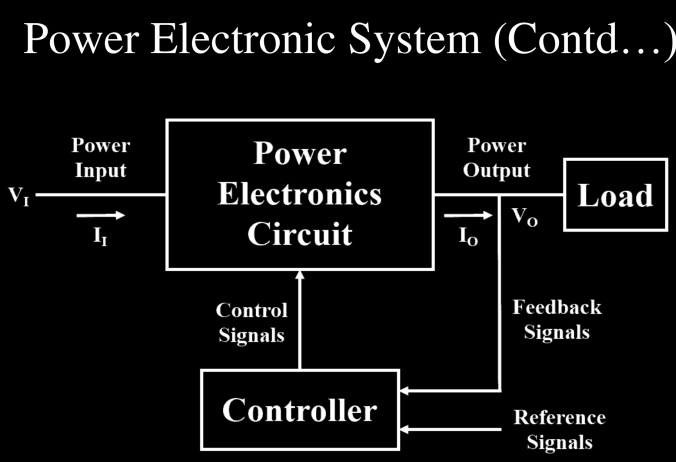


Figure 1: Block diagram of a typical power electronic system showing the power processor, controller, and feedback loop.

Power Electronics vs. Linear / Signal Electronics

The fundamental distinction between linear electronics and power electronics lies in the **operating state of the semiconductor device**.

The Fundamental Theorem of Power Conversion

In a linear regulator or amplifier, the transistor operates continuously in its **active (linear) region**, acting as an electrically controlled variable resistor. The voltage dropped across the transistor is $(V_{\text{in}} - V_{\text{out}})$, and the device dissipates power equal to:

$$P_{\text{loss, linear}} = (V_{\text{in}} - V_{\text{out}}) \cdot I_{\text{load}} \tag{1}$$

Consequently, the maximum theoretical efficiency is capped by the voltage conversion ratio $\eta \leq V_{\text{out}}/V_{\text{in}}$. For instance, stepping down 24 V to 5 V at 10 A yields an efficiency of only $\frac{5}{24} = 20.8\%$, dissipating a staggering 190 W of heat for just 50 W of output power!

In **Power Electronics**, semiconductor devices are strictly operated as **switches**—residing exclusively in two extreme states:

1. **Fully ON (Conduction State):** High load current flows, but the voltage drop across the switch is near zero ($v_s \approx 0$). Hence, instantaneous power dissipation $p(t) = v_s \cdot i \approx 0$.

2. **Fully OFF (Blocking State):** High supply voltage appears across the switch terminals, but the leakage current is near zero ($i_s \approx 0$). Hence, $p(t) = v_s \cdot i \approx 0$.

By routing power through pulsed switching and averaging via lossless reactive filters (L and C), practical efficiencies of $85\% - 99\%$ (typically above 90%) are achieved regardless of the voltage ratio.

Table 1: Comprehensive Comparison: Linear Electronics vs. Power Electronics

Parameter	Linear / Signal Electronics	Power Electronics
Device Operating Mode	Active / Linear region (variable resistance)	Switching mode: Fully ON (Saturation) or Fully OFF (Cut-off)
Power Conversion Efficiency	Poor ($20\% - 60\%$), highly load-dependent	Exceptional ($85\% - 99\%$)
Power Dissipation	Very high; energy difference converted directly to heat	Very low; confined to conduction drop and switching transitions
Power Handling Capacity	Milliwatts to tens of Watts	Watts to Gigawatts ($>$ 1000 MW in HVDC)
Thermal Management	Small PCB copper pours or small clip-on heatsinks	Massive extruded aluminum heatsinks, heat pipes, liquid cold plates
Output Filtering	Linear filtering or direct DC regulation	High-frequency LC low-pass filters
Harmonics & EMI	Pristine low noise, zero switching ripple	High di/dt and dv/dt generate severe harmonics and EMI
Physical Size & Weight	Bulky and heavy for power conversion due to heat	Compact and lightweight due to high switching frequencies (f_{sw})
Primary Design Focus	Signal fidelity, bandwidth, linearity, low distortion	Conversion efficiency, power density, thermal reliability

Power Semiconductor Device Ratings & Frequency Comparison

Understanding the comparative voltage, current, and switching frequency capabilities of different power devices is crucial for converter design (frequently tested exam question). The following table summarizes maximum ratings for standard commercially available devices:

Table 2: Comparative Voltage, Current, and Maximum Switching Frequency Ratings of Power Semiconductor Devices (Source: CET Lecture Slides)

Device Type	Max Voltage (V)	Max Current (A)	Max Switching Frequency (kHz)
Power Diodes	3000	3500	1
Thyristor (SCR)[†]	6000	3500	1
Power MOSFETs	1000	50	100
IGBTs	1200	400	20
GTOs	4000	3000	10
TRIAC	1200	300	0.4
SIT (Static Induction Transistor)	1200	300	100
SITH (Static Induction Thyristor)	4000	2000	20

[†]Row added for completeness (converter-grade SCR, typical values; not on the original slide). *Reading this table:* Table 2 lists typical limits of single *discrete* devices as given in the lecture slides. Table 7 (end of this module) quotes the highest values available from *modules and wide-bandgap parts* (e.g. 6500 V IGBTs, 3000 A modules, 1 MHz MOSFETs), so the two tables intentionally differ. Use Table 2 for slide-style exam answers.

Key Observations from Device Ratings

- **Voltage-Current Trade-Off:** Thyristor-based devices (SCR, GTO, SITH) dominate high-power applications (>1 MW) due to superior voltage (up to 6-8 kV) and current (>3000 A) ratings.
- **Switching Speed vs. Power Capability:** Power MOSFETs offer the highest switching frequencies (100 kHz+) but are limited to lower voltages ($\leq$ 1000 V) and currents ($\leq$ 50 A for single die).
- **IGBTs: The Workhorse of Modern Power Electronics:** IGBTs combine the gate-drive simplicity of MOSFETs with the high-current capability of BJTs, making them the dominant choice for medium-power applications (1 kW - 1 MW).
- **Application Matching:** Low-voltage, high-frequency SMPS ($<$ 100 V, $>$ 100 kHz): MOSFETs or GaN FETs. Medium-voltage motor drives (600 V, 10 $-$ 20 kHz): IGBTs. High-voltage traction/HVDC (>3 kV): Press-pack IGBTs or GTOs.

Ideal vs. Practical Power Switches

To rigorously analyze converter circuits, we first establish the baseline properties of a hypothetical **ideal switch** and compare them against the non-idealities of **practical solid-state switches**.

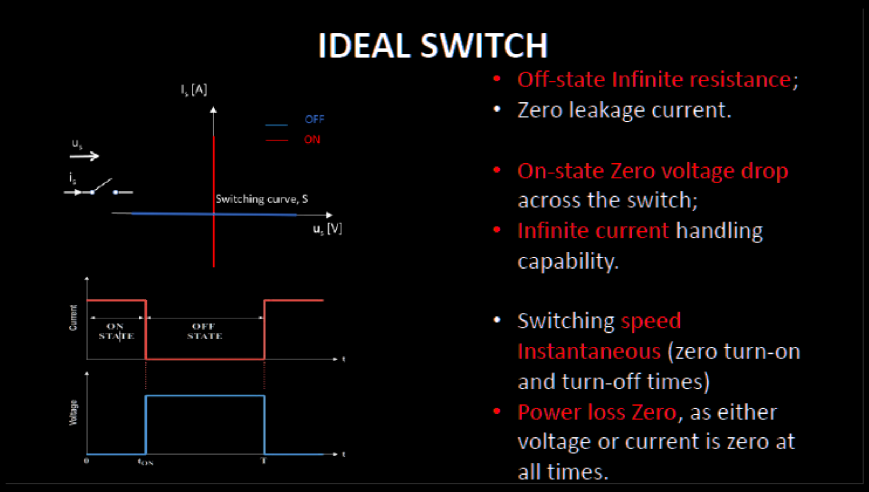


Figure 2: Characteristics of an ideal power electronic switch: zero on-state voltage drop, infinite off-state resistance, instantaneous switching, and zero power loss. (Source: CET Lecture Slides)

Characteristics of an Ideal Switch

An ideal power switch acts as a perfect electrical conductor when closed and a perfect insulator when open:

1. **Zero On-State Voltage Drop:** $V_{\text{on}} = 0$ at any current level; the on-state resistance $R_{\text{on}} = 0$.
2. **Infinite Current Carrying Capacity:** Can carry unlimited forward and reverse current without damage.
3. **Infinite Off-State Resistance:** Zero off-state leakage current ($I_{\text{leak}} = 0$) under infinite blocking voltage.
4. **Infinite Breakdown Voltage:** Capable of blocking infinite forward and reverse voltages ($V_{BR} = \infty$).
5. **Instantaneous Switching:** Transition times are zero ($t_{\text{on}} = 0, t_{\text{off}} = 0$). The rate of change of current (di/dt) and voltage (dv/dt) can be infinitely large.
6. **Zero Control / Drive Power:** Zero energy required to trigger or maintain the switch in the ON or OFF state.
7. **Complete Stability:** Operating parameters are completely independent of junction temperature and age.

Non-Idealities of Practical Solid-State Switches

Real semiconductor devices deviate from the ideal switch in several critical dimensions:

1. **Finite On-State Drop (V_{on}):** Practical devices exhibit an on-state threshold knee voltage (V_0) and dynamic internal resistance (r_d or $R_{DS(\text{on})}$). This gives rise to continuous **conduction losses**.
2. **Finite Leakage Current (I_{leak}):** In the blocking state, thermally generated minority carriers create a small reverse leakage current (1 μA to several mA), producing **off-state losses**.
3. **Finite Switching Times (t_r, t_f):** Carrier recombination and internal junction capacitances prevent instantaneous transitions. During turn-on and turn-off, voltage and current overlap simultaneously, generating massive peak power pulses and resulting in **switching losses**.
4. **Finite Voltage and Current Limits:** Every device has strict maximum ratings ($V_{RRM}, V_{DSS}, I_{C,\text{max}}$). Exceeding these causes immediate avalanche breakdown or thermal destruction.
5. **Stress Limits (di/dt and dv/dt):** High di/dt causes localized current crowding and burnout; high dv/dt triggers capacitive displacement currents that cause spurious gate turn-on. External protection (snubbers) is mandatory.
6. **Finite Control Power:** Gate charges must be shuttled at high frequencies, demanding finite drive currents ($I_{G,\text{peak}} = \Delta V_G / R_G$) and gate drive power.

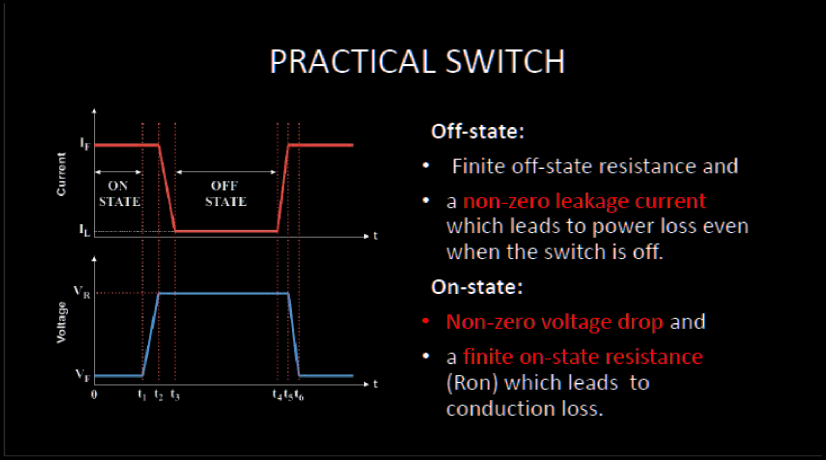


Figure 3: Characteristics of a practical power electronic switch showing non-zero on-state voltage drop, finite off-state leakage current, and finite switching times leading to conduction and switching losses. (Source: CET Lecture Slides)

Switch Classification Taxonomy

Power semiconductor devices are categorized along three fundamental operational axes:

1. **Controllability Axis:**
   - **Uncontrolled Switches (Power Diodes):** The switch state (ON or OFF) is determined entirely by the external power circuit polarity. No gate terminal exists.
   - **Semi-Controlled Switches (SCR / Thyristor):** Can be turned ON by an external gate current pulse when forward-biased, but **cannot be turned OFF by gate control**. Conduction is terminated only when the power circuit forces anode current below the holding current (I_H) through natural or forced commutation.

- **Fully-Controlled Switches (BJT, MOSFET, IGBT, GTO):** Can be turned ON and turned OFF on demand via continuous or pulsed gate/base control signals.

2. **Voltage Withstand Capability:**
   - **Unipolar Voltage Withstand:** Can block voltage of only one polarity (e.g., Power MOSFETs and modern punch-through IGBTs block only positive forward voltage; reverse voltage is clamped by the body diode or causes breakdown).
   - **Bipolar Voltage Withstand:** Can block both large positive and large negative voltages (e.g., conventional SCRs and *symmetrical* GTOs; note that the common *asymmetrical* GTOs block only a few tens of volts in reverse and need an anti-parallel diode).
3. **Current Conduction Capability:**
   - **Unidirectional Current:** Conduction is restricted to a single direction (e.g., Diodes, SCRs, BJTs, IGBTs).
   - **Bidirectional Current:** Can conduct in both directions (e.g., TRIACs, reverse-conducting thyristors (RCT), or Power MOSFETs utilizing third-quadrant reverse channel conduction).

Analytical Derivation of Power Losses in Power Switches

Total energy dissipation in a power semiconductor device over a switching period $T_s = 1/f_{sw}$ comprises the following components:

$$P_{\text{total}} = P_{\text{cond}} + P_{\text{sw}} + P_{\text{oss}} + P_{\text{leak}} + P_{\text{gate}} \tag{2}$$

The first four are dissipated *inside* the switch and set the heat-sink size. The gate-drive loss $P_{\text{gate}} = Q_g\, V_{GS}\, f_{sw}$ is dissipated in the driver and gate resistor, not in the die, so it is excluded from the junction-temperature calculation (it matters for driver sizing, Section on gate drives).

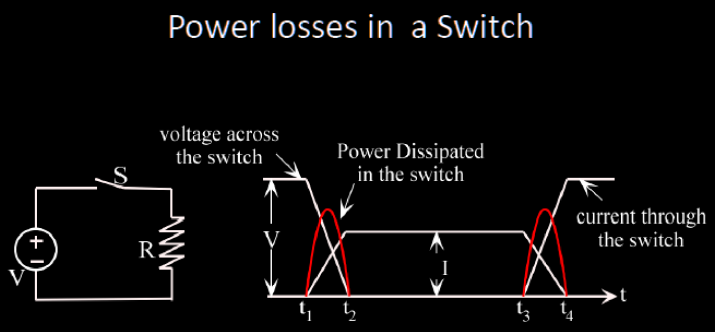


Figure 4: Dynamic switching waveforms and power loss pulses across a practical power switch during turn-on (t_1 to t_2, P_{on}) and turn-off (t_3 to t_4, P_{off}) transitions under clamped inductive switching. (Source: CET Lecture Slides)

1. Conduction Loss (P_{cond})

During the on-state interval $t_{cond} = D \cdot T_s$ (where D is the duty cycle), the switch carries load current $i(t)$. Practical devices are modeled as a threshold drop V_0 in series with an on-state resistance r_d:

$$v_s(t) = V_0 + r_d \cdot i(t) \tag{3}$$

The instantaneous power is $p(t) = v_s(t) \cdot i(t) = V_0 i(t) + r_d i^2(t)$. Integrating over one full cycle:

$$P_{cond} = \frac{1}{T_s} \int_0^{DT_s} \left[V_0 i(t) + r_d i^2(t) \right] dt \tag{4}$$

Master Conduction Loss Formula

$$P_{cond} = V_0 \cdot I_{avg} + r_d \cdot I_{rms}^2 \tag{5}$$

where I_{avg} is the average switch current and I_{rms} is the root-mean-square current through the device during period T_s.

- **For Power MOSFETs:** $V_0 \approx 0$, hence conduction loss is purely resistive:

$$P_{cond,\text{MOSFET}} = I_{D,rms}^2 \cdot R_{DS(on)} \tag{6}$$

- **For IGBTs & Diodes:** Threshold voltage dominates ($V_0 = V_{CE(sat)0} \approx 0.7 - 1.2$ V), giving a quasi-linear relationship with the *average* current through the device, $I_{C,avg}$ (already averaged over the full period T_s, so the duty cycle is *not* multiplied in again):

$$P_{cond,\text{IGBT}} \approx V_{CE(sat)} \cdot I_{C,avg} = V_{CE(sat)} \cdot D \cdot I_{C,on} \quad \text{(flat-top on-state current } I_{C,on}\text{)} \tag{7}$$

2. Switching Loss (P_{sw})

During each transition the switch carries current and blocks voltage at the same time. How the two waveforms overlap decides the loss coefficient, so two cases must be kept apart.

Case A: Voltage and current cross simultaneously (resistive load). Let both transition linearly over the same interval t_r:

$$i(t) = I_L \left(\frac{t}{t_r} \right), \quad v(t) = V_d \left(1 - \frac{t}{t_r} \right) \quad \text{for } 0 \le t \le t_r \tag{8}$$

$$E_{on} = \int_0^{t_r} v(t) \cdot i(t)\, dt = V_d I_L \int_0^{t_r} \left(\frac{t}{t_r} - \frac{t^2}{t_r^2} \right) dt = V_d I_L \left[\frac{t_r}{2} - \frac{t_r}{3} \right] = \frac{1}{6} V_d I_L t_r \tag{9}$$

Case B: Clamped inductive switching (choppers, inverters, converters with a freewheeling diode). This is the practical case. The freewheeling diode holds the switch voltage at V_d while the current rises, and the inductor holds the current at I_L while the voltage falls, so the two waveforms *do not* cross simultaneously. At turn-on, current rises $0 \to I_L$ over t_{ri} with $v = V_d$; then voltage falls $V_d \to 0$ over t_{fv} with $i = I_L$. Each interval is a triangle of height $V_d I_L$:

$$E_{on} = \underbrace{\tfrac{1}{2} V_d I_L\, t_{ri}}_{\text{current rise}} + \underbrace{\tfrac{1}{2} V_d I_L\, t_{fv}}_{\text{voltage fall}} = \frac{1}{2} V_d I_L\, t_{c(on)}, \qquad t_{c(on)} = t_{ri} + t_{fv} \tag{10}$$

Turn-off is the mirror image (voltage rises over t_{rv} at $i = I_L$, then current falls over t_{fi} at $v = V_d$):

$$E_{off} = \frac{1}{2} V_d I_L\, t_{c(off)}, \qquad t_{c(off)} = t_{rv} + t_{fi} \tag{11}$$

Here $t_{c(on)}$ and $t_{c(off)}$ are the total voltage–current *crossover* intervals. For the same transition time, hard inductive switching loses $3\times$ more than the resistive Case A ($\frac{1}{2}$ vs. $\frac{1}{6}$). Multiplying by the switching frequency f_{sw}:

Switching Power Loss Equations (Clamped Inductive Load)

$$P_{on} = \frac{1}{2} V_d I_L\, t_{c(on)}\, f_{sw}, \qquad P_{off} = \frac{1}{2} V_d I_L\, t_{c(off)}\, f_{sw} \tag{12}$$

$$P_{sw} = P_{on} + P_{off} = \frac{1}{2} V_d \cdot I_L \cdot \left(t_{c(on)} + t_{c(off)}\right) \cdot f_{sw} = \left(E_{on} + E_{off}\right) f_{sw} \tag{13}$$

When a datasheet quotes only a current rise time t_r and fall time t_f, the common exam shortcut is $P_{sw} \approx \frac{1}{2} V_d I_L (t_r + t_f) f_{sw}$, i.e. treating t_r and t_f as the crossover intervals. This slightly under-estimates the loss because it ignores t_{fv} and t_{rv}; use the datasheet E_{on}, E_{off} curves when they are given.

3. Output Capacitive Loss (P_{oss})

Power MOSFETs and IGBTs have an intrinsic output capacitance $C_{oss} = C_{ds} + C_{gd}$. During the off-state, this capacitor charges to V_d, storing energy $E_C = \frac{1}{2} C_{oss} V_d^2$. At turn-on, this energy is dissipated directly inside the conducting channel:

$$P_{oss} = \frac{1}{2} C_{oss} \cdot V_d^2 \cdot f_{sw} \tag{14}$$

4. Off-State Leakage Loss (P_{leak})

During the blocking period $(1 - D)T_s$, a small leakage current I_{leak} flows under blocking voltage V_d:

$$P_{leak} = V_d \cdot I_{leak} \cdot (1 - D) \tag{15}$$

Because I_{leak} is typically < 1 mA, P_{leak} represents less than 0.1% of total losses and is neglected in thermal calculations unless operating near maximum junction temperature limits.

Crucial Design Trade-Off: Frequency vs. Efficiency

Conduction loss is **independent of switching frequency** (f_{sw}), whereas switching loss scales **linearly with** f_{sw}.

- Increasing f_{sw} shrinks inductors ($L \propto 1/f_{sw}$) and capacitors ($C \propto 1/f_{sw}$), reducing converter weight and volume.
- However, as f_{sw} exceeds a critical threshold, switching losses dominate, causing temperature runaway unless bulky heat sinks are added.
- Wide Bandgap (SiC/GaN) switches push this frequency barrier upward by reducing transition times (t_r, t_f) by a factor of 10.

Thermal Modeling & Heat Sink Sizing

All electrical losses generated within the microscopic semiconductor die ($1 - 100$ mm^2) are converted into heat. This heat must be transferred to the ambient environment to keep the junction temperature T_j below the maximum permissible operating limit ($T_{j,\max} \approx 125°\text{C} - 150°\text{C}$ for Silicon, $175°\text{C} - 200°\text{C}$ for SiC). Exceeding $T_{j,\max}$ causes thermal breakdown, dopant migration, and device destruction.

The Thermal-Electrical Analogy

One-dimensional steady-state heat conduction is governed by Fourier's law, which is mathematically identical to Ohm's law.

Steady-State Equivalent Thermal Circuit

In a practical power semiconductor package (such as TO-220, TO-247, or press-pack pucks), heat flows through three distinct physical layers in series:

The thermal parameters are defined as:

Table 3: Direct Correspondence Between Electrical and Thermal Domains

Thermal Quantity	Symbol & Unit	Electrical Quantity	Symbol & Unit
Temperature Difference	$\Delta T = T_1 - T_2$ [°C or K]	Potential Difference (Voltage)	$V = V_1 - V_2$ [V]
Heat Flow Rate (Power)	P_D [W or J/s]	Electric Current	I [A or C/s]
Thermal Resistance	R_θ or θ [°C/W or K/W]	Electrical Resistance	R [Ω or V/A]
Thermal Capacitance	C_θ [J/°C]	Electrical Capacitance	C [F or C/V]
Ohm's Law	$\Delta T = P_D \cdot R_\theta$	Ohm's Law	$V = I \cdot R$

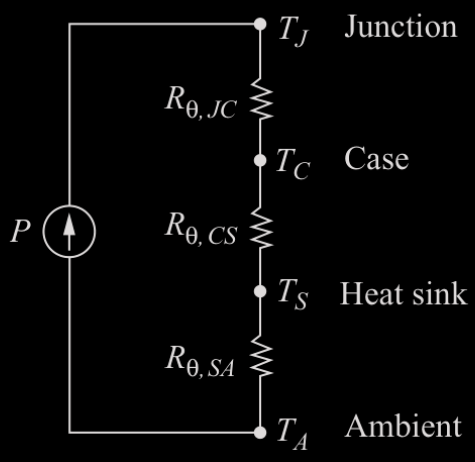


Figure 10.22 The electrical circuit equivalent for a transistor mounted to a heat sink.

Figure 5: One-dimensional steady-state electrical equivalent thermal resistance circuit for a semiconductor device mounted on a heat sink, showing junction-to-case ($R_{\theta,JC}$), case-to-sink ($R_{\theta,CS}$), and sink-to-ambient ($R_{\theta,SA}$) thermal resistances. (Source: Daniel W. Hart, Fig. 10.22)

- $R_{\theta,jc}$ (**Junction-to-Case Thermal Resistance**): Internal resistance from the silicon die to the metal backing tab of the package. Fixed by manufacturer package geometry and bonding materials ($0.2 - 2.0$°C/W).
- $R_{\theta,cs}$ (**Case-to-Sink Thermal Resistance**): Interface resistance across the mechanical mounting boundary. Controlled by surface roughness, contact pressure, and thermal interface material (TIM):
  - Dry metal-to-metal contact: $R_{\theta,cs} \approx 0.5 - 1.0$°C/W (microscopic air voids act as thermal insulators with $k_{\text{air}} \approx 0.026$ W/m · K).
  - With Zinc Oxide / Silicone thermal grease: $R_{\theta,cs} \approx 0.1 - 0.2$°C/W.
  - With electrical insulating mica or Sil-Pad washer: $R_{\theta,cs} \approx 0.3 - 0.5$°C/W.
- $R_{\theta,sa}$ (**Sink-to-Ambient Thermal Resistance**): Resistance from the heat sink surfaces to the ambient air stream via conduction, convection, and radiation.

Fundamental Steady-State Thermal Equations

The total junction-to-ambient thermal resistance $R_{\theta,ja}$ is the simple series sum:

$$R_{\theta,ja} = R_{\theta,jc} + R_{\theta,cs} + R_{\theta,sa} \tag{16}$$

The steady-state temperature rise at the junction is:

$$T_j - T_a = P_D \cdot R_{\theta,ja} = P_D \cdot (R_{\theta,jc} + R_{\theta,cs} + R_{\theta,sa}) \tag{17}$$

The intermediate case and sink temperatures are given by:

$$T_c = T_j - P_D \cdot R_{\theta,jc} = T_a + P_D \cdot (R_{\theta,cs} + R_{\theta,sa}) \tag{18}$$

$$T_s = T_c - P_D \cdot R_{\theta,cs} = T_a + P_D \cdot R_{\theta,sa} \tag{19}$$

Heat Sink Sizing Design Procedure

In converter design, $T_{j,\max}$, T_a, P_D, $R_{\theta,jc}$, and $R_{\theta,cs}$ are known constraints. The design objective is to calculate the **maximum permissible thermal resistance of the heat sink** ($R_{\theta,sa}$):

Heat Sink Sizing Equation

$$R_{\theta,sa} \leq \frac{T_{j,\max} - T_a}{P_D} - (R_{\theta,jc} + R_{\theta,cs}) \tag{20}$$

Design Warning: Operating Without a Heat Sink

If a power switch is operated without a heat sink, the case is exposed directly to ambient air, governed by the case-to-ambient resistance $R_{\theta,ca}$. Because $R_{\theta,ca}$ is typically very large ($40 - 65$°C/W for a standard TO-220 package):

$$P_{D,\max(\text{no sink})} = \frac{T_{j,\max} - T_a}{R_{\theta,jc} + R_{\theta,ca}} \approx \frac{150 - 40}{1.5 + 50} \approx 2.13 \text{ W} \tag{21}$$

Even though a TO-220 MOSFET may be rated for 100 A and 150 W on its datasheet, it **cannot dissipate more than ~ 2 W in free air** without a heat sink before incinerating!

Transient Thermal Impedance ($Z_\theta(t)$)

Under pulsed or intermittent operation (e.g., motor starting surges, short-circuit faults), the thermal mass of the silicon chip, copper package, and aluminum heat sink absorbs heat. The dynamic thermal behavior is governed by the **thermal capacitance** $C_\theta = m \cdot c_p$ (where m is mass and c_p is specific heat capacity).

The time-dependent temperature rise is characterized by the **transient thermal impedance**:

$$Z_{\theta,jc}(t) = \frac{T_j(t) - T_c}{P_D} = R_{\theta,jc}\left(1 - e^{-t/\tau_{th}}\right) \tag{22}$$

where $\tau_{th} = R_\theta C_\theta$ is the thermal time constant.

- Silicon die thermal time constant: $\tau_{die} \approx 1$ ms $- 10$ ms.
- Device package / case time constant: $\tau_{case} \approx 0.5$ s $- 2$ s.
- Extruded aluminum heat sink time constant: $\tau_{sink} \approx 100$ s $- 1000$ s.

Because of these large thermal capacitances, semiconductor switches can tolerate heavy momentary overcurrents (5 to 10 times rated current) for milliseconds without exceeding $T_{j,\max}$, an insight critical for sizing semiconductor protective fuses.

Uncontrolled Power Switches: Power Diodes

Physical Structure and Conductivity Modulation

Conventional low-power signal diodes consist of a simple $p - n$ junction. However, an abrupt $p - n$ junction cannot withstand the high reverse breakdown voltages ($>$ 100 V to several kV) required in power electronics without becoming excessively resistive in the forward direction.

To overcome this fundamental physical limitation, the **Power Diode** incorporates a three-layer structure: $p^+ - n^- - n^+$.

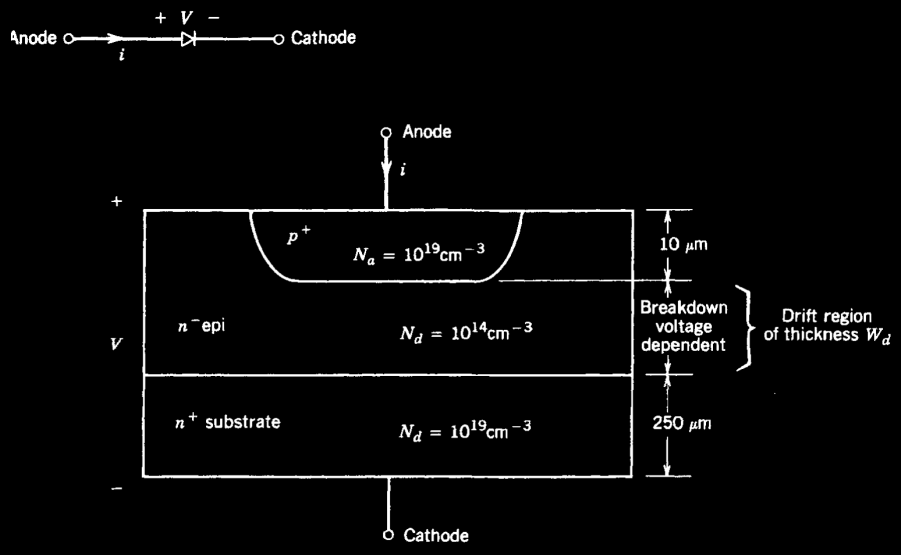


Figure 6: Cross-sectional physical structure of a power diode showing the heavily doped p^+ anode emitter, lightly doped n^- drift region (breakdown voltage dependent thickness W_d), and heavily doped n^+ cathode substrate. (Source: Ned Mohan, Fig. 20-1)

Role of the n^- Drift Region

The lightly doped middle layer is known as the **drift region**.

- **Reverse Voltage Blocking:** When reverse-biased ($V_A < V_K$), the junction between p^+ and n^- is reverse-biased. Because the n^- layer is lightly doped ($N_d \ll N_a$), the depletion region extends almost entirely into the n^- drift layer.
- The reverse breakdown voltage V_{BR} is directly related to the width and doping concentration of the drift region. Supporting high breakdown voltages requires a **wide drift layer** and **extremely light doping**.
- **Design Trade-Off:** A thicker drift region increases the voltage blocking capability but also increases the on-state resistance. Power diode design involves optimizing this trade-off between blocking voltage and forward conduction loss.

Punch-Through vs. Non-Punch-Through Diodes

Punch-Through (PT) Diodes: The drift layer width is designed such that under maximum rated reverse voltage, the depletion region extends all the way through the n^- layer and reaches the n^+ substrate. This allows a thinner drift region for a given voltage rating, reducing on-state resistance and improving forward conduction characteristics.

Non-Punch-Through (NPT) Diodes: The depletion region does not extend through the entire drift region even at maximum rated voltage. These devices have higher blocking voltage margin but slightly higher on-state voltage drop.

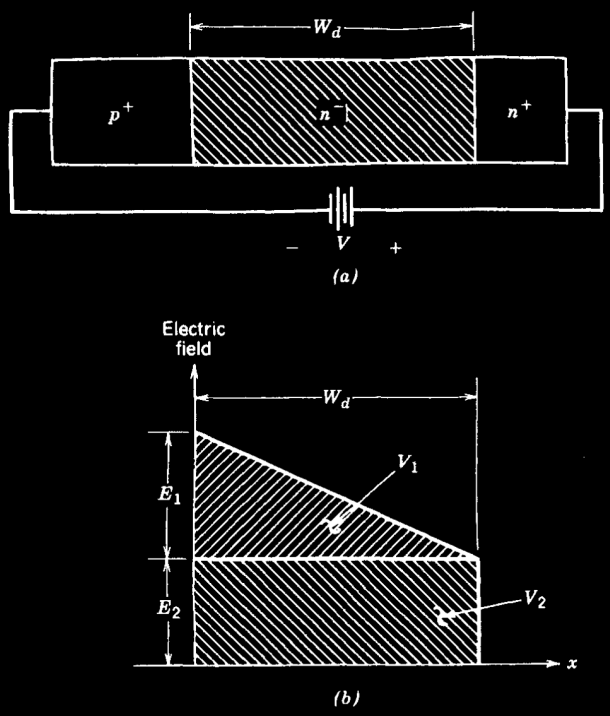


Figure 7: Punch-through condition in a reverse-biased power diode: (a) depletion layer extending across the entire n^- drift region; (b) electric field profile showing trapezoidal distribution with $V_{BR} = \frac{1}{2}(E_1 + E_2) \cdot W_d$. (Source: Mohan, Fig. 20-3)

Conductivity Modulation: The Secret to Low Forward Drop

If the drift region remained unmodulated during forward conduction, its ohmic resistance would be catastrophic:

$$R_{drift} = \frac{\rho \cdot W_d}{A} = \frac{W_d}{q\mu_n N_d A} \tag{23}$$

For a 1200 V device, R_{drift} would be several ohms, producing thousands of watts of loss at normal operating currents.

However, when forward-biased ($V_A > V_K$), **high-level double injection** occurs:

1. The p^+ anode injects an immense concentration of holes into the n^- drift region.
2. Simultaneously, the n^+ cathode injects an equal concentration of electrons to maintain charge neutrality.
3. The injected excess carrier density $\Delta n = \Delta p \sim 10^{16} - 10^{17}$ cm^{-3} exceeds the background doping $N_d \sim 10^{14}$ cm^{-3} (as in Fig. 6) by **two to three orders of magnitude**!

This massive flood of electron-hole plasma causes **conductivity modulation**:

$$\sigma_{on} = q(\mu_n + \mu_p) \cdot \Delta n \gg \sigma_{drift} \tag{24}$$

The effective drift resistance collapses from several ohms to milliohms, enabling the power diode to conduct hundreds of amperes with a forward voltage drop of merely $V_F \approx 0.9 - 1.3$ V.

Classification of Power Diodes

Power diodes are categorized into three distinct classes based on their manufacturing process, recovery characteristics, and target applications:

Table 4: Comprehensive Comparison of Power Diode Classes

Parameter / Feature	Line-Frequency Diodes	Fast Recovery Diodes (FRD)	Schottky Barrier Diodes (SBD)
Internal Junction	$p^+ - n^- - n^+$ Silicon	$p^+ - n^- - n^+$ (Lifetime killed)	Metal-Semiconductor (PtSi/Si)
Carrier Mechanism	Bipolar (Minority + Majority)	Bipolar (Reduced lifetime)	Unipolar (Majority electrons only)
Reverse Breakdown (V_{RRM})	Up to 5000 V	Up to 3000 V	Limited: $\leq 150 - 200$ V (Si)
Forward Current ($I_{F(AV)}$)	Up to 5000 A	Up to 1000 A	Up to 400 A
Forward Voltage Drop (V_F)	$0.8 - 1.2$ V	$1.2 - 2.2$ V	Very low: $0.3 - 0.5$ V
Reverse Recovery Time (t_{rr})	Very slow: $10 - 50\,\mu$s	Ultra-fast: 25 ns $- 1\,\mu$s	Negligible: ≈ 0 ns (zero stored charge)
Reverse Leakage (I_R)	Very low ($\sim \mu$A)	Moderate ($\sim 10\,\mu$A)	High ($\sim$ mA), temp-sensitive
Manufacturing Method	Standard diffusion/alloying	Gold/Platinum doping or electron beam irradiation	Metallization on n-epitaxial silicon
Typical Applications	50/60 Hz mains rectifiers, electroplating, traction	Inverters, SMPS, choppers, freewheeling paths	Low-voltage SMPS (5V, 12V), OR-ing diodes, solar bypass

Schottky Diodes: Why Zero Minority Carrier Storage?

In a standard $p - n$ diode, forward conduction is maintained by minority carriers that diffuse across the junction and must be removed before the diode can block reverse voltage.
A **Schottky Barrier Diode** consists of a metal (such as platinum silicide, molybdenum, or titanium) bonded directly to an n-type semiconductor. Forward current is conducted entirely by **majority carriers** (electrons flowing from the semiconductor into the metal conduction band via thermionic emission).

- Because there are **no minority holes** injected into the semiconductor, there is **zero stored minority charge** ($Q_{RR} \approx 0$).
- Reverse recovery time is determined purely by the discharge of the tiny metal-semiconductor barrier capacitance C_j, allowing switching frequencies well into the megahertz range.
- The low built-in potential barrier gives a forward voltage drop of just $0.3 - 0.4$ V, making Schottky diodes unbeatable for low-voltage, high-efficiency output rectification.

Dynamic Switching Characteristics: The Reverse Recovery Transient

When a conducting diode carrying forward current I_F is commutated OFF by applying a reverse voltage, it does not turn off instantaneously. The dense cloud of electron-hole plasma stored in the drift region during conductivity modulation must be completely evacuated before a depletion region can form.

The reverse recovery process divides into two distinct physical phases:

1. **Storage Interval ($t_a = t_1 - t_0$):**
   - At $t = t_0$, the diode current crosses zero and reverses direction with a rate of fall di/dt determined by external loop inductance (L) and supply voltage ($di/dt = V_R/L$).
   - During t_a, the excess carriers stored near the $p^+ - n^-$ junction are swept out by the reverse current.
   - The diode remains heavily populated with plasma; therefore, the junction voltage drop remains small ($v_D \approx 0$).
   - At $t = t_1$, the carrier concentration at the junction drops to zero, and the reverse current peaks at **Peak Reverse Recovery Current (I_{RR})**.

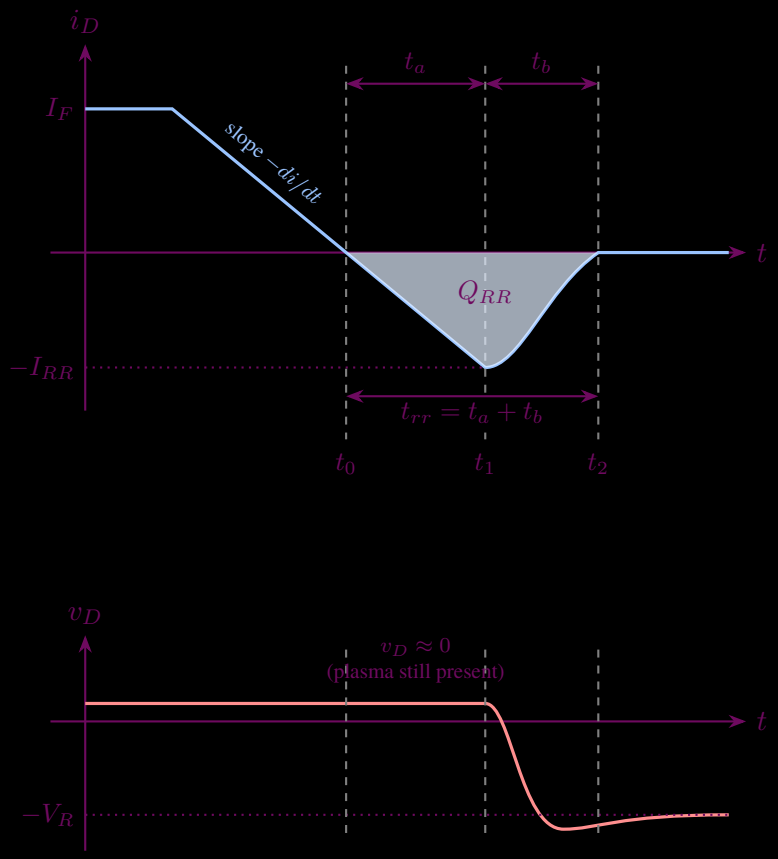


Figure 8: Diode dynamic reverse recovery process showing the forward current commutation, storage interval t_a, fall interval t_b, peak reverse current I_{RR}, stored recovery charge Q_{RR}, and total recovery time $t_{rr} = t_a + t_b$ (redrawn; the diode voltage stays near zero during t_a and rises to $-V_R$ during t_b).

2. **Recombination / Fall Interval ($t_b = t_2 - t_1$):**

   - At t_1, a space-charge depletion layer begins to establish itself across the junction.
   - The reverse current decays from $-I_{RR}$ back to the tiny reverse leakage level as remaining trapped carriers in the bulk drift layer recombine.
   - Simultaneously, the reverse voltage across the diode rises from zero to the DC rail voltage V_R.

Mathematical Derivation of t_{rr} and I_{RR}

Approximating the reverse recovery current waveform as a triangle:

$$Q_{RR} = \frac{1}{2} \cdot I_{RR} \cdot t_{rr} = \frac{1}{2} \cdot I_{RR} \cdot (t_a + t_b) \tag{25}$$

During interval t_a, the current changes from 0 to I_{RR} at rate di/dt:

$$I_{RR} = \left(\frac{di}{dt}\right) \cdot t_a \tag{26}$$

Define the softness factor $S = t_b/t_a$ (introduced below), so that $t_{rr} = t_a + t_b = (1 + S)\, t_a$. Substituting $I_{RR} = (di/dt)\, t_a$ into the triangle area:

$$Q_{RR} = \frac{1}{2} I_{RR}\, t_{rr} = \frac{1}{2} \left(\frac{di}{dt}\right) t_a \cdot (1 + S)\, t_a = \frac{1}{2}(1 + S) \left(\frac{di}{dt}\right) t_a^2 \tag{27}$$

Solving for t_a and then forming t_{rr} and I_{RR}:

Master Reverse Recovery Equations (general S)

$$t_a = \sqrt{\frac{2Q_{RR}}{(1+S)\,(di/dt)}}, \qquad t_{rr} = (1+S)\,t_a = \sqrt{\frac{2Q_{RR}\,(1+S)}{di/dt}} \tag{28}$$

$$I_{RR} = \left(\frac{di}{dt}\right) t_a = \sqrt{\frac{2Q_{RR}\,(di/dt)}{1+S}} \tag{29}$$

Snappy diode ($S \to 0$, so $t_{rr} \approx t_a$): $t_{rr} \approx \sqrt{2Q_{RR}/(di/dt)}$ and $I_{RR} \approx \sqrt{2Q_{RR}\,(di/dt)}$.
Soft diode ($S = 1$): $t_{rr} = \sqrt{4Q_{RR}/(di/dt)}$ and $I_{RR} = \sqrt{Q_{RR}\,(di/dt)}$.

Note that for a given Q_{RR} and di/dt, a softer diode (larger S) has a *longer* t_{rr} but a *smaller* peak I_{RR}, which is exactly why it is gentler on the circuit.

The Softness Factor (S) and Snappy Recovery Dangers

The ratio of the decay interval t_b to the storage interval t_a is defined as the **Softness Factor** (S):

$$S = \frac{t_b}{t_a} \tag{30}$$

- **Soft Recovery Diode ($S \geq 1$):** The reverse current decays gradually back to zero. The rate of fall during t_b ($|di/dt|_{t_b}$) is small, producing negligible inductive voltage overshoot.
- **Snappy / Hard Recovery Diode ($S \ll 1$, typically $S \leq 0.2$):** The current snaps abruptly to zero from $-I_{RR}$ in a few nanoseconds.

Consequences of Snappy Recovery

In any physical circuit, stray layout inductance L_σ is unavoidable ($10 - 50$ nH in PCB traces and busbars). During the snappy interval t_b, the rapid current truncation generates a severe inductive voltage spike across the diode:

$$v_{\text{spike}} = V_R + L_\sigma \cdot \left|\frac{di}{dt}\right|_{t_b} \tag{31}$$

This voltage spike can exceed the diode's reverse breakdown rating V_{RRM}, triggering catastrophic avalanche destruction. Furthermore, snappy recovery excites high-frequency ringing between L_σ and diode junction capacitance C_j, radiating broadband electromagnetic interference (EMI).

Power Losses Caused by Reverse Recovery

During interval t_b, reverse voltage $v_D(t)$ and reverse current $i_D(t)$ exist simultaneously across the diode. Assuming a linear rise of voltage to V_R and linear decay of current from I_{RR}, the energy dissipated in the diode per switching cycle is:

$$E_{rr} = \int_{t_a}^{t_{rr}} v_D(t) \cdot i_D(t)\,dt \approx \frac{1}{6} V_R \cdot I_{RR} \cdot t_b = \frac{1}{6} V_R \cdot I_{RR} \cdot t_{rr} \left(\frac{S}{1+S}\right) \tag{32}$$

$$P_{rr} = E_{rr} \cdot f_{sw} \tag{33}$$

Transistor Turn-On Stress: When the freewheeling diode recovers, the complementary active switch (e.g., MOSFET or IGBT) must carry the load current I_L **plus the diode's reverse recovery current** I_{RR}:

$$I_{\text{switch, peak}} = I_L + I_{RR} \tag{34}$$

This drastically increases the turn-on switching loss (E_{on}) of the transistor, which often dwarfs the loss dissipated in the diode itself!

Power Diode Ratings

1. V_{RRM} **(Repetitive Peak Reverse Voltage):** Maximum instantaneous reverse voltage that the diode can withstand repeatedly without avalanche breakdown.

2. V_{RSM} **(Non-Repetitive Peak Surge Reverse Voltage):** Maximum transient reverse overvoltage spike (duration < 10 ms) permitted under fault conditions ($V_{RSM} \approx 1.1 - 1.2 \times V_{RRM}$).

3. $I_{F(AV)}$ **(Maximum Average Forward Current):** Maximum allowable half-sine or rectangular average forward current based on maximum junction temperature $T_{j,\text{max}}$ and internal package resistance.

4. $I_{F(RMS)}$ **(RMS Forward Current):** Critical for sizing conductors and estimating resistive I^2R heating in package leads.

5. I_{FSM} **(Peak One-Cycle Non-Repetitive Surge Current):** Maximum allowable half-sine wave current surge (50 Hz, 10 ms) that the diode can survive under a short-circuit fault without exploding.

6. I^2t **Rating (Thermal Capacity for Fuse Coordination):** Measure of the energy capacity of the semiconductor die during short-circuit faults ($t < 10$ ms):

$$I^2t = \int_0^{t_f} i^2(t)\,dt \tag{35}$$

   For safe protection, the clearing I^2t of the fast-acting semiconductor fuse must be strictly less than the I^2t rating of the diode: $(I^2t)_{\text{fuse}} < (I^2t)_{\text{diode}}$.

Semi-Controlled Switches: Thyristors (SCR)

Physical Structure, Symbol, and Operational Modes

The **Silicon Controlled Rectifier (SCR)**, or simply **Thyristor**, is a four-layer, three-junction, three-terminal semiconductor device that acts as a bistable switch. It is classified as a **semi-controlled switch** because it can be turned ON at will via a low-power gate current pulse, but **cannot be turned OFF via the gate**. Turn-off requires the external circuit to reduce the anode current below a critical threshold (the holding current) and maintain reverse bias for a sufficient duration.

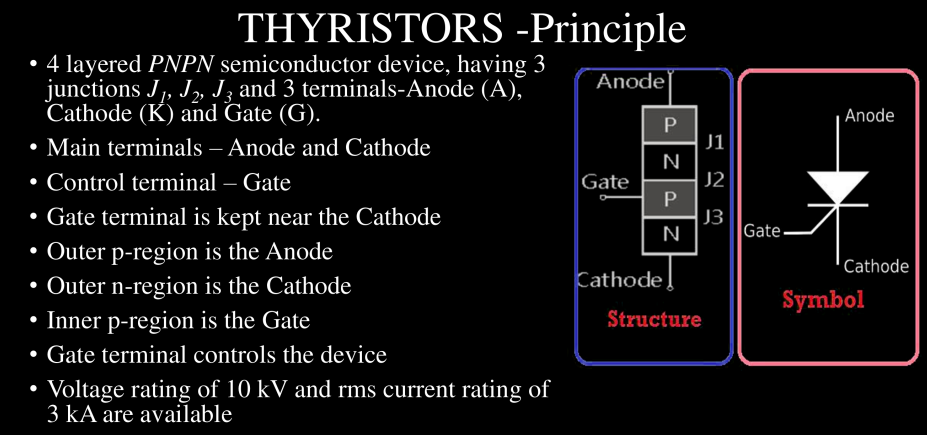


Figure 9: Internal four-layer PNPN physical structure of an SCR alongside its standard electrical circuit symbol. (Source: CET Lecture Slides)

The four semiconductor layers form three internal $p - n$ junctions:

- J_1: Between p_1 and n_1 (Outer anode junction).
- J_2: Between n_1 and p_2 (Central blocking junction).
- J_3: Between p_2 and n_2 (Outer cathode/gate junction).

The lightly doped n_1 drift layer is wide to support high forward and reverse blocking voltages (up to $6 - 8$ kV).

The Five SCR Turn-On Methods

Five Methods of Thyristor Turn-On (Frequently Asked 3-Mark/9-Mark Question)

An SCR can be triggered into conduction by five distinct physical mechanisms:

1. **Forward Voltage Triggering (Breakover):**
   - When forward anode-cathode voltage exceeds the **forward breakover voltage** (V_{BO}), avalanche multiplication occurs across the reverse-biased central junction J_2.
   - Avalanche-generated carriers initiate regenerative feedback, causing spontaneous turn-on without any gate signal.
   - This method is generally avoided in controlled applications as it bypasses gate control and can lead to uncontrolled switching.
2. **Gate Triggering (Normal Method):**
   - This is the **standard and most widely used triggering method** in practical power electronic circuits.
   - A positive gate current pulse ($I_G > I_{GT}$) is injected into the gate terminal, which injects electrons across junction J_3 into the p_2 base layer.
   - This initiates carrier multiplication across J_2, collapsing the blocking junction and latching the SCR into forward conduction.
   - Provides precise control over the instant of turn-on (firing angle control).
3. **Temperature Triggering (Thermal Runaway):**
   - At elevated junction temperatures, thermally generated minority carrier concentration increases exponentially: $n_i \propto e^{-E_g/2kT}$.
   - This increases the reverse leakage currents I_{CBO1} and I_{CBO2} in the two-transistor model.
   - If temperature rises sufficiently, leakage current amplified by regenerative gain can exceed the threshold for latch-up, causing spontaneous unwanted turn-on.
   - This is an **undesirable failure mode** that must be prevented by adequate thermal management and derating.
4. **Light Triggering (Photothyristors / Light-Activated SCR):**
   - When high-intensity light (typically infrared or laser) is directed onto the central blocking junction J_2, photon absorption generates electron-hole pairs directly in the depletion region.
   - These photogenerated carriers provide the initial seed current to initiate regenerative turn-on.
   - Used in specialized **optically isolated gate drive circuits** for high-voltage applications (HVDC converters) where electrical isolation of several kilovolts is required between control circuitry and power stages.
5. dv/dt **Triggering (Capacitive Displacement Current):**

- The reverse-biased central junction J_2 has an intrinsic junction capacitance C_{j2}.
- When a rapidly rising forward voltage is applied across the anode-cathode (dv/dt is very high), a displacement charging current flows: $i_c = C_{j2}\frac{dv}{dt}$.
- This displacement current flows through the cathode junction J_3, acting exactly like an externally injected gate current.
- If i_c exceeds the gate trigger current threshold (I_{GT}), the SCR **turns ON spuriously without any deliberate gate signal**.
- **Prevention:** An $R_s - C_s$ snubber circuit connected in parallel with the SCR limits the rate of rise of voltage and prevents false triggering.

SCR Voltage Ratings: Standard Notation Definitions

Power semiconductor datasheets specify multiple distinct voltage ratings using standardized subscript notation. Understanding these definitions is critical for safe device selection (frequently tested exam question):

Table 5: SCR Voltage Rating Definitions with Standard Subscript Notation (Source: CET Lecture Slides)

Rating Symbol	Definition
V_{DWM}	**Peak Working Forward Blocking Voltage:** Maximum instantaneous forward voltage that the SCR can withstand continuously during normal repetitive operation without triggering or breakdown. Design circuits to keep peak voltage below this value.
V_{DRM}	**Peak Repetitive Forward Blocking Voltage:** Maximum forward voltage that the device can block repetitively (on every cycle) without damage. Typically, $V_{DRM} \approx 1.1 \times V_{DWM}$.
V_{DSM}	**Peak Non-Repetitive (Surge) Forward Blocking Voltage:** Maximum transient forward voltage overshoot that the SCR can withstand for a very short duration (typically one half-cycle, 10 ms at 50 Hz) without permanent damage. Typically, $V_{DSM} \approx 1.2 - 1.3 \times V_{DRM}$. This rating accounts for occasional lightning surges or switching transients.
V_{RWM}	**Peak Working Reverse Blocking Voltage:** Maximum instantaneous reverse voltage (cathode positive with respect to anode) that the SCR can withstand continuously during normal repetitive operation.
V_{RRM}	**Peak Repetitive Reverse Blocking Voltage:** Maximum reverse voltage that the device can block repetitively without avalanche breakdown. Typically, $V_{RRM} \approx 1.1 \times V_{RWM}$.
V_{RSM}	**Peak Non-Repetitive (Surge) Reverse Blocking Voltage:** Maximum transient reverse voltage spike that the SCR can withstand for a very short duration without permanent damage. Typically, $V_{RSM} \approx 1.2 - 1.3 \times V_{RRM}$.

Design Safety Margins

In practical converter design, the actual peak circuit voltage should be kept well below the device working voltage ratings to account for transient overshoots, temperature variations, and aging:

$$V_{\text{circuit, peak}} \leq 0.7 - 0.8 \times V_{DWM} \quad \text{(Forward)} \qquad V_{\text{circuit, peak}} \leq 0.7 - 0.8 \times V_{RWM} \quad \text{(Reverse)} \quad (36)$$

A safety margin of 20-30% is standard industry practice.

Static V-I Characteristics

Depending on the terminal voltage polarity and gate signal, the SCR operates in three distinct quadrants/modes:

1. **Reverse Blocking Mode ($V_{AK} < 0$):**

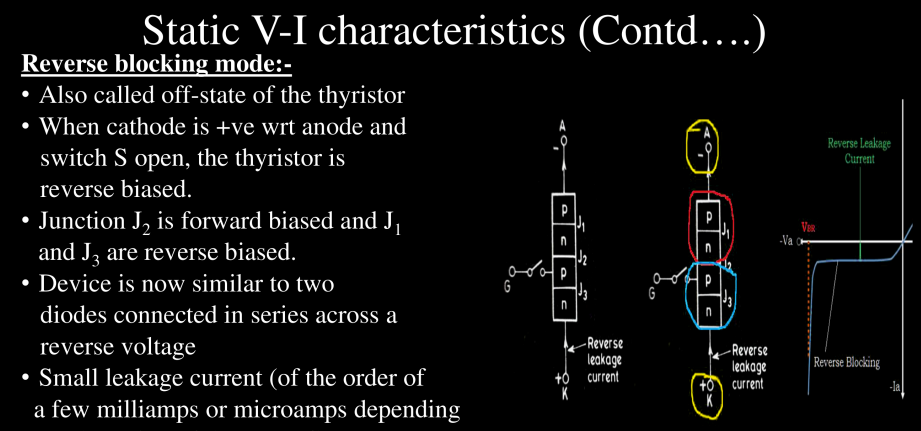


Figure 10: Static V-I characteristics of an SCR illustrating forward blocking, forward breakover (V_{BO}), forward conduction, latching current (I_L), holding current (I_H), gate current effect, and reverse breakdown (V_{BR}). (Source: CET Lecture Slides)

- The cathode is positive relative to the anode.
- Junctions J_1 and J_3 are reverse-biased; central junction J_2 is forward-biased.
- The device behaves like two series reverse-biased diodes. A tiny reverse leakage current flows until the **Reverse Breakdown Voltage** (V_{BR}) is reached, whereupon avalanche multiplication occurs, destroying the device if current is not externally limited.

2. **Forward Blocking Mode** ($V_{AK} > 0, I_G = 0$)**:**
   - The anode is positive relative to the cathode, but no gate signal is applied.
   - Junctions J_1 and J_3 are forward-biased, but the central junction J_2 **is reverse-biased**.
   - The entire forward voltage appears across the depletion region of J_2. A tiny forward leakage current flows. The SCR acts as an open switch until the **Forward Breakover Voltage** (V_{BO}) is exceeded.

3. **Forward Conduction Mode** ($V_{AK} > 0, I_G > 0$)**:**
   - When a gate current pulse is injected into p_2, electrons are injected across J_3, initiating regenerative carrier multiplication across J_2.
   - The central junction J_2 collapses into heavy conductivity modulation; all three junctions become forward-biased.
   - The voltage across the SCR drops from hundreds of volts to the forward on-state drop:

$$V_T \approx 1.0 - 1.5 \text{ V} \tag{37}$$

   The anode current is now limited entirely by the external load impedance.

Latching Current (I_L) vs. Holding Current (I_H)

These two critical current thresholds are frequently tested in engineering examinations:

- **Latching Current (I_L):** The **minimum anode current** that the SCR must reach **at the time of turn-on** so that the device remains in the forward conduction state even after the gate trigger pulse is completely removed.
  - Associated with the **turn-on process**.
  - If the gate pulse terminates before $i_A(t)$ reaches I_L, the regenerative loop cannot sustain itself and

the SCR reverts immediately to forward blocking.

- **Holding Current (I_H):** The **minimum anode current** below which the conducting SCR drops out of forward conduction and returns to the forward blocking state.
  - Associated with the **turn-off (commutation) process**.
  - To turn off an SCR, the anode current must be forced below I_H for a duration exceeding the turn-off time t_q.
- **Fundamental Inequality:**

$$I_L > I_H \qquad (\text{Typically, } I_L \approx 2 \text{ to } 3 \times I_H) \tag{38}$$

Two-Transistor Analogy of SCR

The regenerative latch-up physics of the SCR can be mathematically analyzed by slicing the four-layer $p_1 - n_1 - p_2 - n_2$ structure into two interconnected, complementary bipolar junction transistors:

- Transistor Q_1 (**PNP**): Formed by p_1 (emitter), n_1 (base), and p_2 (collector).
- Transistor Q_2 (**NPN**): Formed by n_1 (collector), p_2 (base), and n_2 (emitter).

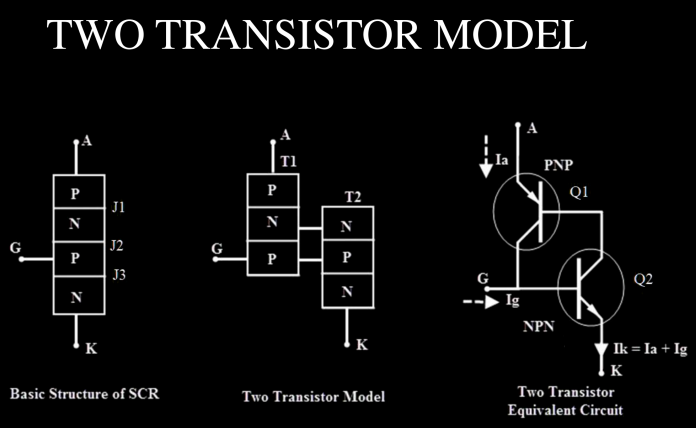


Figure 11: The Two-Transistor regenerative model of the Silicon Controlled Rectifier: decomposing the four-layer PNPN structure into complementary PNP (Q_1) and NPN (Q_2) transistors. (Source: CET Lecture Slides)

Step-by-Step Mathematical Derivation of Anode Current

Let:

- α_1: Common-base current gain of transistor Q_1 (I_{C1}/I_{E1}).
- α_2: Common-base current gain of transistor Q_2 (I_{C2}/I_{E2}).
- I_{CBO1}: Collector-base reverse saturation leakage current of Q_1.
- I_{CBO2}: Collector-base reverse saturation leakage current of Q_2.

From standard bipolar junction transistor theory, the collector current for each transistor is:

$$I_{C1} = \alpha_1 I_{E1} + I_{CBO1} = \alpha_1 I_A + I_{CBO1} \tag{39}$$

$$I_{C2} = \alpha_2 I_{E2} + I_{CBO2} = \alpha_2 I_K + I_{CBO2} \tag{40}$$

From Kirchhoff's Current Law applied to the external terminals of the SCR:

$$I_K = I_A + I_G \tag{41}$$

Substituting Eq. (41) into Eq. (40):

$$I_{C2} = \alpha_2 (I_A + I_G) + I_{CBO2} \tag{42}$$

Across the central reverse-biased junction J_2, the total anode current passing through the device must equal the sum of the two collector currents:

$$I_A = I_{C1} + I_{C2} \tag{43}$$

Substituting Eqs. (39) and (42) into Eq. (43):

$$I_A = (\alpha_1 I_A + I_{CBO1}) + [\alpha_2 (I_A + I_G) + I_{CBO2}] \tag{44}$$

Grouping all terms containing I_A on the left-hand side:

$$I_A - \alpha_1 I_A - \alpha_2 I_A = \alpha_2 I_G + I_{CBO1} + I_{CBO2} \tag{45}$$

$$I_A \left[1 - (\alpha_1 + \alpha_2)\right] = \alpha_2 I_G + I_{CBO1} + I_{CBO2} \tag{46}$$

Master SCR Anode Current Equation

$$I_A = \frac{\alpha_2 I_G + I_{CBO1} + I_{CBO2}}{1 - (\alpha_1 + \alpha_2)} \tag{47}$$

Physical Mechanism of Regenerative Turn-On

The common-base current gain α is a strong function of emitter current I_E:

- At low currents (blocking state), carrier recombination in the depletion layers is high, resulting in very low gains: $\alpha_1 \ll 1$ and $\alpha_2 \ll 1$. Thus, $\alpha_1 + \alpha_2 \ll 1$.
- With zero gate current ($I_G = 0$), the anode current is merely the microscopic leakage current:

$$I_A = \frac{I_{CBO1} + I_{CBO2}}{1 - (\alpha_1 + \alpha_2)} \approx I_{CBO1} + I_{CBO2} \tag{48}$$

- When gate current I_G is injected into the base of Q_2, emitter current $I_{E2} = I_K$ increases.
- This immediately increases α_2, boosting collector current I_{C2}.
- Because I_{C2} is directly wired into the base of Q_1, the emitter current $I_{E1} = I_A$ increases, which in turn boosts α_1 and collector current I_{C1}.
- Current I_{C1} feeds back into the base of Q_2, further increasing I_{C2}.

This establishes a **positive regenerative feedback loop**. As:

$$(\alpha_1 + \alpha_2) \longrightarrow 1 \tag{49}$$

The denominator in Eq. (47) approaches zero:

$$\lim_{(\alpha_1+\alpha_2)\to 1} I_A = \infty \tag{50}$$

The loop gain exceeds unity, both transistors drive each other into hard saturation, and the device latches ON. Once latched, the internal feedback maintains $\alpha_1 + \alpha_2 \approx 1$, meaning **the gate signal I_G can be completely removed without affecting conduction**.

Dynamic Switching Characteristics

Turn-On Phenomena

When a gate pulse is applied, the transition from forward blocking to forward conduction requires a finite duration called the **turn-on time** ($t_{on} = 1 - 4\,\mu s$):

$$t_{on} = t_d + t_r + t_p \tag{51}$$

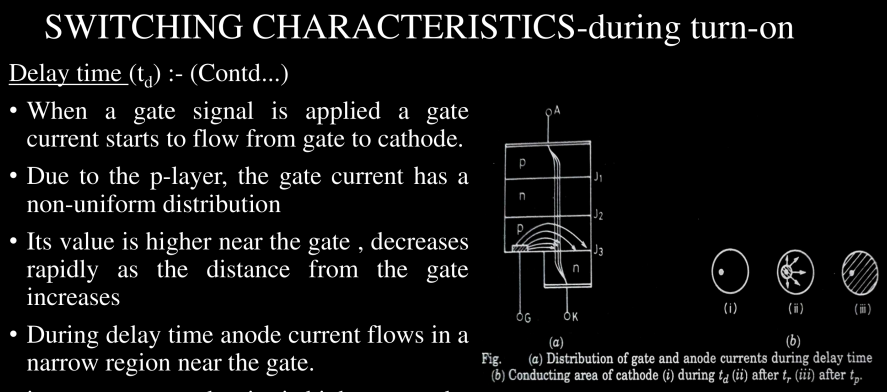


Figure 12: SCR dynamic turn-on switching waveforms showing gate current i_G, anode voltage v_a, anode current i_a, delay time (t_d), rise time (t_r), and spread time (t_p). (Source: CET Lecture Slides)

1. **Delay Time (t_d):** Time elapsed between the arrival of the gate current (10% point) and the instant anode current reaches 10% of its final steady value (I_a). Alternatively, it is the time for anode voltage to fall from V_a to $0.9V_a$. During t_d, gate charge accumulates in the p_2 layer.

2. **Rise Time (t_r):** Time taken for anode current to rise from 10% to 90% of I_a, while anode voltage falls from $0.9V_a$ to $0.1V_a$.

   The di/dt Hazard During Rise Time

   During t_r, the device conducts through a narrow filament directly adjacent to the gate contact. If the rate of rise of anode current (di/dt) is excessively high, the current density in this localized filament becomes astronomical before plasma spreading can occur. This creates severe **localized thermal hot spots** that permanently melt the silicon slice! An external series inductor L_s is mandatory.

3. **Spread Time (t_p):** Time required for conduction to spread from the initial filament across the entire cross-sectional area of the cathode junction. Voltage drops from $0.1V_a$ to steady on-state drop V_T.

Turn-Off Phenomena (Commutation Mechanics)

Once an SCR is conducting, the gate loses all control. To turn the device off, the anode current must be reduced below the holding current I_H, and a reverse voltage must be maintained across the device for a minimum duration known as the **device turn-off time** (t_q):

$$t_q = t_{rr} + t_{gr} \tag{52}$$

1. **Reverse Recovery Time (t_{rr}):** The current reverses at $-di/dt$ to sweep out stored carriers from outer junctions J_1 and J_3. Once swept, J_1 and J_3 regain their reverse dielectric blocking capacity.



Figure 13: SCR dynamic turn-off switching waveforms showing reverse recovery interval (t_{rr}), gate recovery interval (t_{gr}), device turn-off time ($t_q = t_{rr} + t_{gr}$), and circuit turn-off time (t_c). (Source: CET Lecture Slides)

2. **Gate Recovery Time (t_{gr}):** Trapped minority charge carriers at the central junction J_2 cannot be swept out by external reverse voltage because J_2 is forward-biased under reverse terminal polarity! These carriers can disappear **only by natural recombination**. t_{gr} is the time required for recombination to reduce carrier density below the critical threshold.

3. **Device Turn-Off Time (t_q):** The total internal time $t_q = t_{rr} + t_{gr}$ required by the SCR before it is capable of blocking forward voltage without re-triggering ($t_q \approx 50 - 100\,\mu$s for converter-grade; $3 - 20\,\mu$s for inverter-grade).

4. **Circuit Turn-Off Time (t_c):** The time interval during which the external power circuit applies a reverse voltage across the SCR.

The Golden Commutation Rule

For successful turn-off (commutation), the circuit turn-off time must strictly exceed the manufacturer device turn-off time with a safety margin:

$$t_c > t_q \qquad \text{(Safety factor: } t_c \geq 1.5 \text{ to } 2 \times t_q\text{)} \tag{53}$$

If forward voltage reappears across the SCR before t_q has elapsed ($t_c < t_q$), the uncombined carriers at J_2 immediately trigger regenerative latch-up, resulting in catastrophic commutation failure (inverter shoot-through).

SCR Protection Schemes & Snubber Circuit Design

di/dt Protection

As established, localized current crowding during turn-on destroys the cathode junction. To protect against this, an inductor L_s is connected in series with the SCR:

$$L_s \geq \frac{V_m}{(di/dt)_{\text{max}}} \tag{54}$$

where V_m is the peak supply voltage and $(di/dt)_{\text{max}}$ is the device rating.

dv/dt Protection: The Snubber Circuit

Under forward blocking, junction J_2 behaves as a parallel-plate capacitor with junction capacitance C_{j2}. When a rapidly rising forward voltage is applied across the anode-cathode terminals, a capacitive charging displacement current flows:

$$i_c = \frac{dq}{dt} = \frac{d}{dt}(C_{j2}v) = C_{j2}\frac{dv}{dt} + v\frac{dC_{j2}}{dt} \approx C_{j2}\frac{dv}{dt} \tag{55}$$

This displacement current flows directly through the cathode junction J_3, acting exactly like an externally injected gate current! If i_c exceeds the gate trigger current (I_{GT}), the SCR **turns ON spuriously** without any gate pulse.

To limit dv/dt, an $R_s - C_s$ **snubber circuit** is connected in parallel with the SCR.

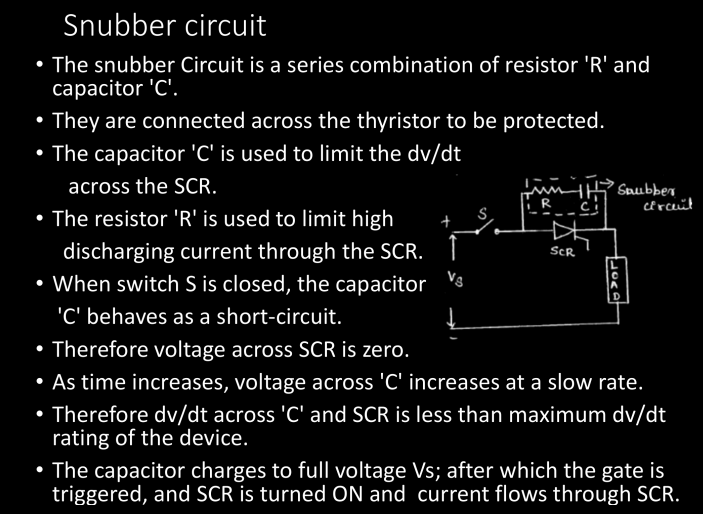


Figure 14: Complete thyristor protection scheme: series inductor L_s for di/dt protection and parallel $R_s - C_s$ snubber network for dv/dt protection. (Source: CET Lecture Slides)

Snubber Circuit Design Equations

When the supply is switched on (a step of peak value V_m) while the SCR is still blocking, the source drives the loop inductance L_s in series with the snubber branch R_s–C_s (which sits across the SCR). Writing $i = C_s\, dv_C/dt$, the capacitor voltage obeys the second-order equation

$$L_s C_s \frac{d^2 v_C}{dt^2} + R_s C_s \frac{dv_C}{dt} + v_C = V_m \tag{56}$$

The voltage seen by the SCR is the whole branch voltage, $v_T = v_C + R_s i = V_m - L_s\, di/dt$. At $t = 0^+$ the inductor current is zero and $v_C = 0$, so $v_T(0) = 0$ and the initial slope follows from $di/dt = V_m/L_s$:

$$\left(\frac{dv_T}{dt}\right)_{t=0} = R_s \left.\frac{di}{dt}\right|_0 = \frac{V_m\, R_s}{L_s} \tag{57}$$

This is the steepest slope the SCR experiences (for $\xi \approx 0.65$ a numerical simulation of the Problem 5 circuit confirms that later slopes never exceed it). Note that it is the slope across the *SCR* (the R_s–C_s branch), not across C_s alone, whose own slope is zero at $t = 0$. Hence the dv/dt rating requires

$$R_s \le \frac{L_s\,(dv/dt)_{\max}}{V_m} \tag{58}$$

To avoid ringing that would push the SCR voltage towards $2V_m$ (as $\xi \to 0$), the loop is designed for a damping ratio $\xi \approx 0.65$:

$$\xi = \frac{R_s}{2}\sqrt{\frac{C_s}{L_s}} \approx 0.65 \quad \Longrightarrow \quad C_s = L_s\left(\frac{2\xi}{R_s}\right)^2 \tag{59}$$

(The capacitor alone overshoots by $e^{-\pi\xi/\sqrt{1-\xi^2}} \approx 7\%$ at $\xi = 0.65$; the SCR voltage also includes the $R_s i$ drop, so its overshoot is larger, about 20% in the Problem 5 numbers. Allow for this when choosing V_{DRM}.)

Finally, when the SCR fires, C_s discharges through R_s and the SCR. Limiting this discharge current to the SCR's repetitive peak rating $I_{T,\text{max}}$ gives

$$R_s \geq \frac{V_m}{I_{T,\text{max}}} \tag{60}$$

The two limits on R_s can be met only if $L_s \geq V_m^2/\left[I_{T,\text{max}}(dv/dt)_{\text{max}}\right]$; if the L_s chosen for di/dt is smaller than this, increase L_s.

Snubber Design Procedure

1. $L_s \geq \dfrac{V_m}{(di/dt)_{\text{max}}}$, and also $L_s \geq \dfrac{V_m^2}{I_{T,\text{max}}\,(dv/dt)_{\text{max}}}$ (take the larger).
2. Choose R_s within $\dfrac{V_m}{I_{T,\text{max}}} \leq R_s \leq \dfrac{L_s\,(dv/dt)_{\text{max}}}{V_m}$.
3. $C_s = L_s\left(\dfrac{2\xi}{R_s}\right)^2$ with $\xi \approx 0.65$. (If R_s is set at its upper limit: $C_s = \dfrac{4\xi^2 V_m^2}{L_s\,(dv/dt)_{\text{max}}^2} \approx \dfrac{1.69\,V_m^2}{L_s\,(dv/dt)_{\text{max}}^2}$.)
4. $P_{\text{snubber}} \approx C_s V_m^2 f$ (conservative upper estimate; $\frac{1}{2}C_s V_m^2 f$ is dissipated per discharge event, and a chopper snubber loses both its charge and discharge halves).

Quick alternative seen in many textbooks: if L_s is neglected the snubber is an RC charging circuit and $dv/dt \approx 0.632\,V_m/(R_s C_s)$ (average over the first time constant). The procedure above is the one that includes L_s, as required when di/dt protection is present.

Gate Control, Drive Requirements, and Firing Circuits

Gate Pulse Requirements

1. **Minimum Gate Current & Voltage:** Gate signal must exceed the minimum values (V_{GT}, I_{GT}) specified at the lowest operating temperature.
2. **Gate Power Limits:** Instantaneous peak gate power (P_{GM}) and average power ($P_{G(AV)}$) must not exceed datasheet bounds to prevent gate junction burnout.
3. **Pulse Width Requirement with Inductive Load:** When an SCR fires into an $R-L$ load, the anode current rises exponentially:

$$i_A(t) = \frac{V_s}{R}\left(1 - e^{-Rt/L}\right) \approx \frac{V_s}{L}\cdot t \quad (\text{for } t \ll L/R) \tag{61}$$

The gate pulse width t_p must be wide enough for $i_A(t)$ to reach the **latching current** I_L:

$$t_p \geq t_{\text{latch}} = \frac{L \cdot I_L}{V_s} \tag{62}$$

If $t_p < t_{\text{latch}}$, the SCR turns off the instant the gate pulse ends!

Gate V-I Characteristics and Load Line

The gate-cathode junction of an SCR behaves somewhat like a $p-n$ diode but exhibits significant manufacturing spread between individual devices of the same type. The manufacturer provides a boundary region of permissible $V_g - I_g$ characteristics (bounded by max/min limits and average gate power dissipation curve). A load line is drawn from the gate source voltage E_s with slope $-1/R_s$ to determine the operating point.

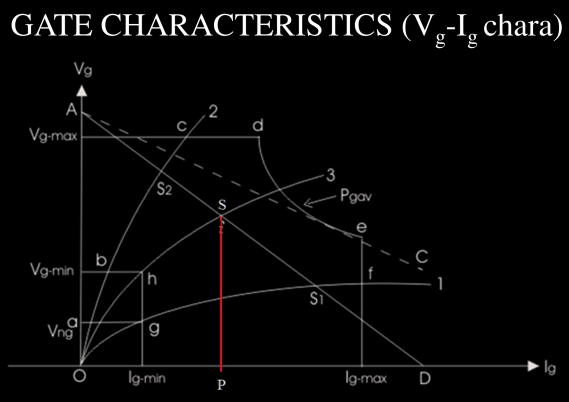


Figure 15: SCR gate V_g-I_g characteristics showing limit curves 1 & 2 (manufacturing spread), non-triggering voltage V_{ng}, average gate power dissipation hyperbola P_{gav}, and the gate load line $E_s = V_g + I_g R_s$. The operating point S must lie within the permissible region bounded by curves 1, 2, and P_{gav}. (Source: CET Lecture Slides)

Resistance (R) Triggering Circuit

Resistance triggering is the simplest and most economical firing method for phase-controlled thyristors. It derives gate current directly from the alternating mains supply voltage ($v_s = V_m \sin \omega t$) through a resistive network.

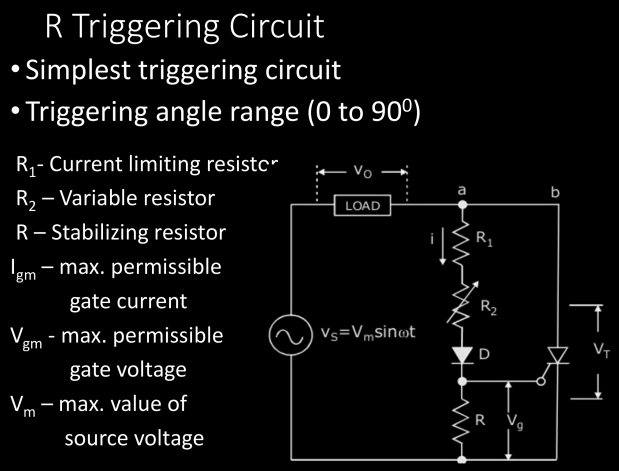


Figure 16: Resistance (R) triggering circuit for SCR phase control. R_1 is a fixed current-limiting resistor, R_2 is a variable potentiometer for adjusting the firing angle α, R is a stabilizing resistor preventing noise trigger, and diode D blocks negative gate voltage. (Source: CET Lecture Slides)

Circuit Operation and Component Functions:

1. **Fixed Resistor R_1 (Gate Protection):** When the variable resistor R_2 is adjusted to zero, the maximum instantaneous gate current must not exceed the manufacturer's peak gate current rating (I_{gm}):

$$R_1 \geq \frac{V_m}{I_{gm}} \tag{63}$$

2. **Variable Resistor R_2 (Firing Angle Control):** Adjusts the slope of gate voltage $v_g(t)$ to control the instant at which v_g reaches the gate trigger voltage V_{gt}. To ensure that triggering can occur at peak supply (V_m):

$$R_1 + R_2 \leq \frac{V_m - V_{gt} - V_D}{I_{gt}} \tag{64}$$

where $V_D \approx 0.7$ V is the forward drop of diode D, and I_{gt} is the minimum gate trigger current.

3. **Diode D (Reverse Gate Protection):** Acts as a half-wave rectifier, allowing current to flow into the gate only during the positive half-cycle of v_s. During the negative half-cycle, D blocks reverse voltage, protecting the reverse-biased gate-cathode junction J_3 from catastrophic breakdown.

4. **Stabilizing Resistor R:** Connected directly between gate and cathode to shunt false thermal leakage currents and high-frequency noise spikes away from J_3, preventing spurious triggering.

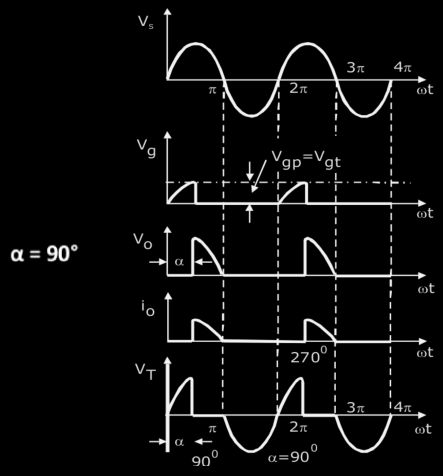


Figure 17: R triggering circuit waveforms for $\alpha = 90°$ ($V_{gp} = V_{gt}$): source voltage v_s, gate voltage v_g, output voltage v_o, output current i_o, and thyristor voltage v_T. (Source: CET Lecture Slides)

Fundamental Limitation of R Triggering: Why $\alpha \leq 90°$?

In a purely resistive circuit, the gate voltage $v_g(t)$ is in phase with the source voltage $v_s(t) = V_m \sin \omega t$. The peak of the gate voltage occurs precisely at $\omega t = 90°$:

- If $v_g(t)$ reaches the required trigger threshold V_{gt} before or at $90°$, the SCR triggers at that instant: $0° \leq \alpha \leq 90°$.
- If R_2 is set so high that v_g does not reach V_{gt} by $\omega t = 90°$, then because $v_s(t)$ decreases after $90°$, $v_g(t)$ also decreases and will **never reach** V_{gt} during the remainder of the cycle.
- Therefore, **the firing angle in an R triggering circuit is strictly restricted to** $0° \leq \alpha \leq 90°$. Firing control in the second quadrant ($90° < \alpha < 180°$) is physically impossible!

Resistance-Capacitance (RC) Triggering Circuit

To overcome the $90°$ firing limit of the pure resistance circuit and achieve smooth firing angle control across nearly the entire positive half-cycle ($0° \leq \alpha \leq 180°$), a capacitor is added to form an **RC phase-shift network**.

Operating Physics and Extended Firing Angle:

1. **Capacitive Phase Lag:** The voltage across capacitor C lags the applied source voltage $v_s(t)$ by an angle $\phi = \arctan(\omega RC) \leq 90°$.

2. Because $v_c(t)$ lags $v_s(t)$, the capacitor voltage continues to rise even after the source voltage passes its peak at $\omega t = 90°$.

3. By adjusting the variable resistor R, the instant at which v_c reaches $V_{gt} + V_{D1}$ can be smoothly delayed beyond $90°$, achieving a controllable firing angle range of:

$$0° \leq \alpha \leq 180° \tag{65}$$

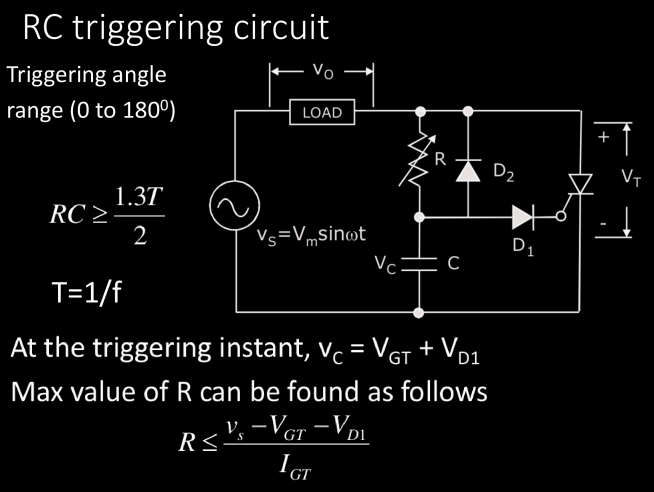


Figure 18: RC triggering circuit for extended firing angle range ($0° \leq \alpha \leq 180°$). Diode D_1 protects the gate against negative bias, while diode D_2 clamps capacitor voltage during the negative half-cycle. (Source: CET Lecture Slides)

4. **Role of Diode D_2 (Negative Half-Cycle Reset):** During the negative half-cycle of supply voltage ($v_s < 0$), diode D_2 becomes forward-biased and conducts, clamping the capacitor voltage to a small negative value:

$$v_c \approx -V_{D2} \approx -0.7 \text{ V} \tag{66}$$

   This resets the capacitor charge and guarantees identical, repeatable initial timing conditions at the beginning of each positive half-cycle.

5. **Design Criterion:** To guarantee that triggering is possible near $180°$, the time constant RC must satisfy:

$$RC \geq \frac{1.3T}{2} = \frac{1.3}{2f} = \frac{1.3\pi}{\omega} \tag{67}$$

   where $T = 1/f$ is the mains period. Furthermore, to supply the required gate trigger current I_{gt}:

$$R \leq \frac{v_s - V_{gt} - V_{D1}}{I_{gt}} \tag{68}$$

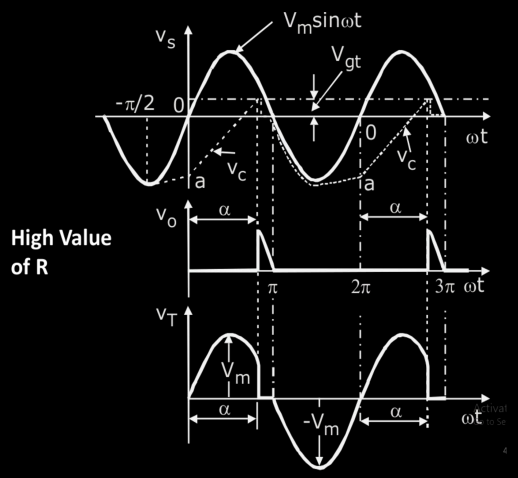


Figure 19: RC triggering waveforms for high value of R (large firing angle): capacitor voltage v_c lags source voltage, producing delayed gate triggering near $\alpha \approx 180°$. (Source: CET Lecture Slides)

UJT Relaxation Oscillator Firing Circuit

While R and RC circuits provide simple phase control, they deliver prolonged gate pulses that cause high average gate power dissipation and lack electrical isolation. The **Uni-Junction Transistor (UJT)** relaxation oscillator generates sharp, high-energy pulses that ensure instantaneous, jitter-free SCR turn-on while minimizing gate losses.

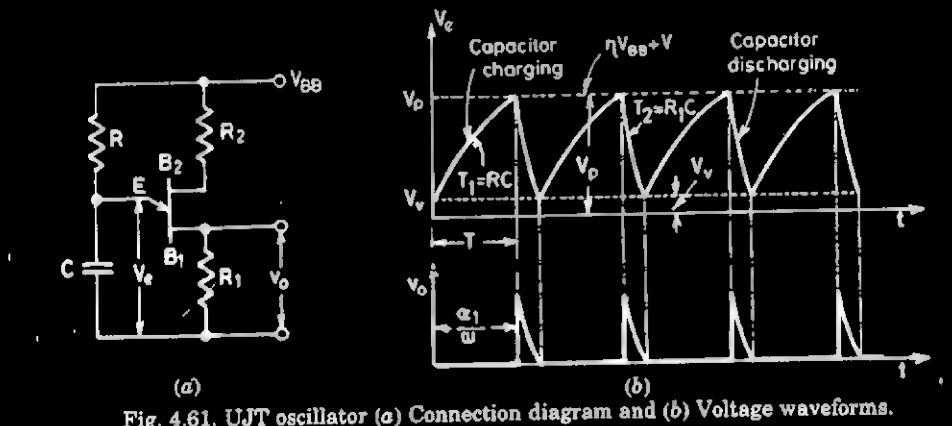


Fig. 4.61. UJT oscillator (a) Connection diagram and (b) Voltage waveforms.

Figure 20: UJT Relaxation Oscillator triggering circuit: (a) connection diagram showing charging network (R_E, C_E), interbase biasing (R_1, R_2), and pulse output to SCR gate; (b) emitter voltage charging waveform $v_E(t)$ between valley point V_V and peak point V_P, alongside the resulting sharp gate trigger pulses $v_{B1}(t)$. (Source: Dr. P. S. Bimbhra, Fig. 4.61)

1. **Intrinsic Standoff Ratio (η):** The internal silicon bar has interbase resistance $R_{BB} = R_{B1}+R_{B2}$. The voltage across R_{B1} before breakdown is ηV_{BB}, where:

$$\eta = \frac{R_{B1}}{R_{B1}+R_{B2}} = \frac{R_{B1}}{R_{BB}} \qquad \text{(Typically, } \eta \approx 0.55 - 0.82\text{)} \tag{69}$$

2. **Peak Point Voltage (V_P):** The emitter junction becomes forward-biased when v_E reaches:

$$V_P = \eta V_{BB} + V_D \tag{70}$$

   where $V_D \approx 0.7$ V is the emitter diode forward drop.

3. **Oscillation Cycle:** Capacitor C_E charges exponentially through R_E toward V_{BB}:

$$v_E(t) = V_{BB}\left(1 - e^{-t/R_E C_E}\right) \tag{71}$$

   When $v_E(t) = V_P$, the UJT enters negative resistance, discharging C_E rapidly through base B_1 and R_1, generating a sharp voltage pulse across R_1 that triggers the SCR. Discharge stops when v_E reaches the valley voltage V_V, and the charging cycle repeats.

4. **Mathematical Derivation of Period T and Frequency f:**

$$V_P = V_V + (V_{BB} - V_V)\left(1 - e^{-T/R_E C_E}\right) \tag{72}$$

   Neglecting V_V and V_D relative to V_{BB}:

$$\eta V_{BB} \approx V_{BB}\left(1 - e^{-T/R_E C_E}\right) \implies e^{-T/R_E C_E} = 1 - \eta \tag{73}$$

UJT Frequency & Design Equations

$$T = R_E C_E \ln\left(\frac{1}{1-\eta}\right), \qquad f = \frac{1}{R_E C_E \ln\left(\frac{1}{1-\eta}\right)} \tag{74}$$

Limits on R_E for Sustained Relaxation Oscillation: To guarantee that the load line intersects the negative resistance region:

$$\frac{V_{BB} - V_P}{I_P} > R_E > \frac{V_{BB} - V_V}{I_V} \tag{75}$$

where I_P is the peak emitter current and I_V is the valley current.

Gate Synchronisation and Isolation

- **AC Line Synchronisation:** To control the firing angle α relative to the AC mains, the DC rail of the UJT circuit (V_{BB}) is derived from the AC supply using a full-wave diode bridge clamped by a Zener diode. At every AC voltage zero-crossing, V_{BB} drops to zero, immediately resetting capacitor C_E and locking the trigger pulse timing to the line frequency.
- **Pulse Transformer Isolation:** High-power SCRs operate at floating line potentials (hundreds of volts), while digital control logic operates at low DC potentials ($3.3\ \text{V} - 5\ \text{V}$). A **pulse transformer** provides galvanic isolation (> 3 kV). To prevent transformer magnetic core saturation, the gate signal is modulated into a train of high-frequency pulses ($10 - 50$ kHz) rather than a continuous DC level.
- **Optocoupler Isolation:** Uses an infrared LED optically coupled across a transparent dielectric barrier to a phototransistor or photo-SCR, offering virtually infinite DC isolation and zero capacitive cross-talk.

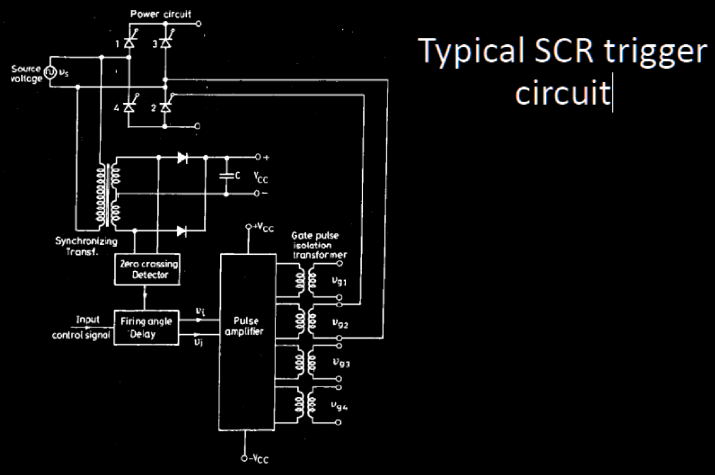


Figure 21: Practical SCR gate trigger system: synchronizing transformer provides zero-crossing reference, firing angle delay block sets α, pulse amplifier drives isolated pulse transformers for multi-SCR bridge firing. (Source: CET Lecture Slides)

Fully-Controlled Switches: Power MOSFETs & IGBTs

Power MOSFETs

The **Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET)** is a unipolar, majority-carrier, voltage-controlled power semiconductor switch. Because conduction is mediated solely by majority electrons (in n-channel devices), power MOSFETs exhibit **zero minority carrier storage delay**, enabling switching speeds in the mega-hertz (> 1 MHz) range.

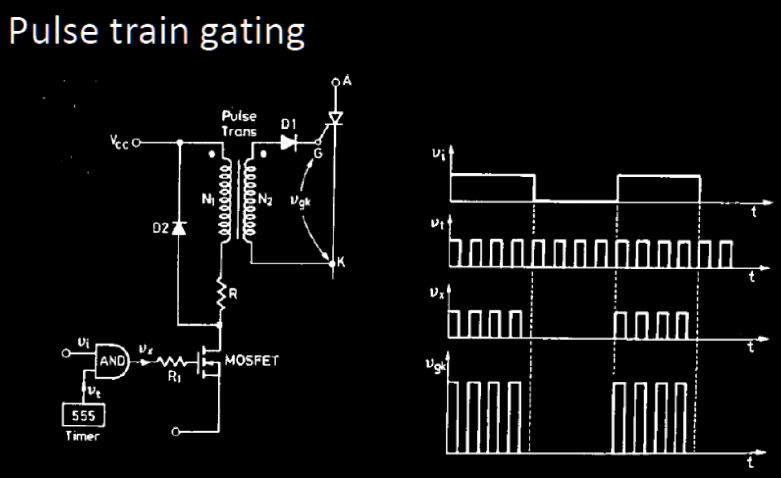


Figure 22: High-frequency carrier (pulse train) gating circuit using a 555 timer and AND gate to generate modulated pulse bursts through a compact pulse transformer, enabling reliable triggering of SCRs with highly inductive loads. (Source: CET Lecture Slides)

Vertical DMOS (VDMOS) Physical Architecture

Planar lateral MOSFETs used in microprocessors cannot withstand high voltages because their breakdown voltage scales with lateral surface distance, consuming vast die area. Modern power MOSFETs utilize a **Vertical Double-Diffused MOS (VDMOS)** architecture:

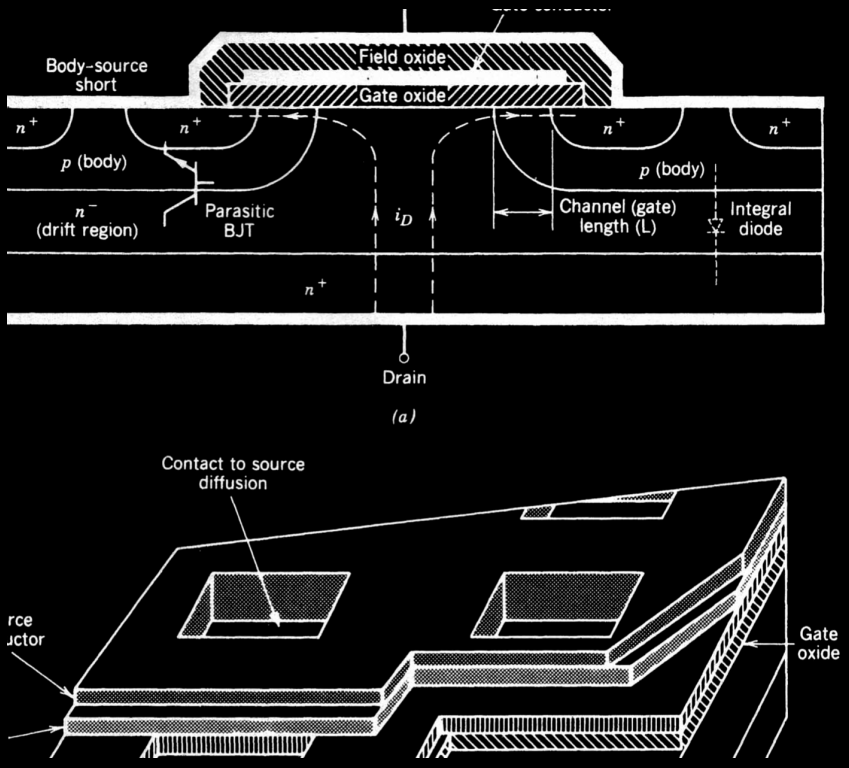


Figure 23: Cross-sectional physical structure of a planar vertical DMOS (VDMOS) power cell showing the inversion channel formation and vertical electron current flow from source to drain. (Source: Ned Mohan, Fig. 22-1)

1. **Vertical Geometry:** The Source contact is on the top surface, while the Drain contact covers the entire bottom substrate. Current flows vertically through the lightly doped n^- drift region. This maximizes cross-sectional conduction area while providing the vertical thickness needed to block high voltages.

2. **Inversion Channel Formation:** Applying a positive gate-to-source voltage V_{GS} exceeding the threshold voltage $V_{th} \approx 2-4$ V repels holes in the p-body surface beneath the gate oxide and attracts electrons. This creates an **inversion n-channel** linking the n^+ source to the n^- drift layer.

3. **Intrinsic Parasitic Body Diode:** The $p-n$ junction between the p-body and n^- drift layer forms an integral anti-parallel body diode directly across the drain-source terminals. It acts as an automatic freewheeling diode in bridge converters, though its reverse recovery is relatively slow in standard silicon MOSFETs.

4. **Parasitic NPN BJT:** An internal BJT is formed by n^+ source (emitter), p-body (base), and n^- drift (collector). To prevent this parasitic BJT from turning on (which causes second breakdown), the source metallization directly shorts the n^+ source and p-body together on the chip surface.

Static Characteristics and On-State Resistance ($R_{DS(on)}$)

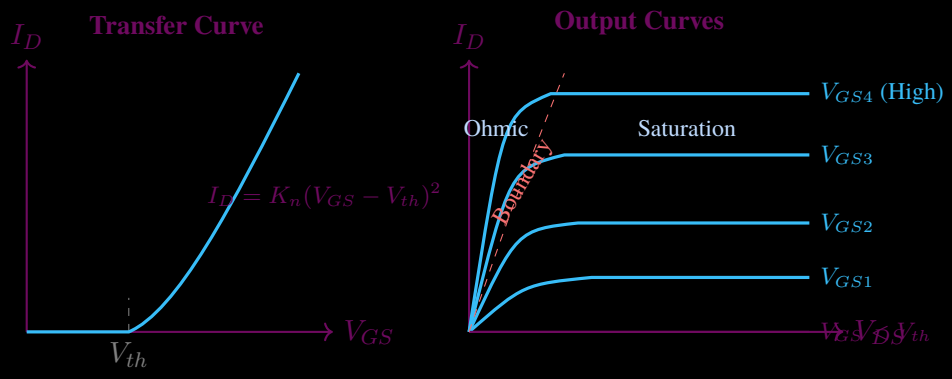


Figure 24: Transfer characteristics (I_D vs. V_{GS}) and output characteristics (I_D vs. V_{DS}) of a power MOSFET.

In the on-state (fully switched ON in the Ohmic/Linear region), the total drain-to-source resistance $R_{DS(on)}$ is the series sum of internal physical components:

$$R_{DS(on)} = R_{\text{source}} + R_{ch} + R_{acc} + R_{JFET} + R_D + R_{sub} \tag{76}$$

- R_{ch}: Inversion channel resistance, inversely proportional to $(V_{GS} - V_{th})$.
- R_{JFET}: Resistance of the neck between adjacent p-bodies, where the vertical current is squeezed; it is reduced by cell-pitch and doping optimisation.
- R_D: **Drift region resistance**. For high-voltage silicon MOSFETs ($V_{BR} > 200$ V), R_D dominates completely:

$$R_D \propto \frac{W_d}{N_d} \propto V_{BR}^{2.5} \tag{77}$$

This $V_{BR}^{2.5}$ exponent (an empirical silicon result; the constant-E_c idealisation of Eq. (84) gives V_{BR}^2, and the extra $V_{BR}^{0.5}$ comes from E_c rising weakly with doping, $E_c \propto N_d^{1/8}$) represents the **Silicon Unipolar Limit**: Doubling the breakdown voltage rating increases $R_{DS(on)}$ by nearly $6\times$! Consequently, high-voltage silicon MOSFETs (> 600 V) have excessively high conduction loss, which led to the development of the IGBT.

Positive Temperature Coefficient: Safe Paralleling

The on-state resistance $R_{DS(on)}$ of a power MOSFET has a **positive temperature coefficient**:

$$\frac{dR_{DS(on)}}{dT_j} > 0 \tag{78}$$

As junction temperature rises, increased lattice vibrations (acoustic phonon scattering) decrease electron mobility μ_n, causing $R_{DS(on)}$ to increase (typically doubling between 25°C and 150°C).

- When MOSFETs are connected in parallel, if one device runs hotter, its $R_{DS(on)}$ increases, which automatically sheds current to cooler parallel devices.
- This **self-balancing current sharing** prevents thermal runaway at normal operating current, so MOSFETs need no external ballast resistors (unlike BJTs).
- **Caveat:** V_{th} has a *negative* temperature coefficient. Below the zero-temperature-coefficient (ZTC) current, a hotter device draws *more* current, so lightly loaded parallel devices can hog current unless the layout is symmetrical and each gate has its own gate resistor. "Easy" therefore means easier than BJTs, not automatic.

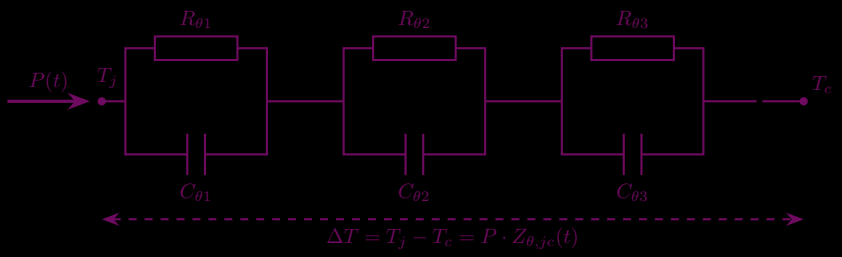


Figure 25: Foster RC-ladder model of the transient thermal impedance from junction to case: each stage is a parallel $R_{\theta i} \parallel C_{\theta i}$ pair with time constant $\tau_i = R_{\theta i} C_{\theta i}$, and the stages are connected in series. The heat flow $P(t)$ plays the role of a current source and ΔT that of a voltage.

For a real package the single exponential of the transient-impedance equation above is extended to n terms, $Z_{\theta,jc}(t) = \sum_{i=1}^{n} R_{\theta i}\left(1 - e^{-t/\tau_i}\right)$ with $\sum_i R_{\theta i} = R_{\theta,jc}$. Each term is one stage of the ladder in Fig. 25, and the datasheet $Z_\theta(t)$ curve is a fit of this sum.

Dynamic Switching Dynamics & The Miller Plateau

Power MOSFET switching speed is governed by the charging and discharging of internal inter-electrode parasitic capacitances:

- C_{gs} (Gate-to-Source capacitance): Constant, governed by gate oxide area.
- C_{gd} (Gate-to-Drain / Miller capacitance): Highly non-linear; increases by up to $100\times$ as V_{DS} drops near zero.
- C_{ds} (Drain-to-Source capacitance): Voltage-dependent junction depletion capacitance.

Datasheets specify: $C_{iss} = C_{gs} + C_{gd}$, $C_{oss} = C_{ds} + C_{gd}$, $C_{rss} = C_{gd}$.

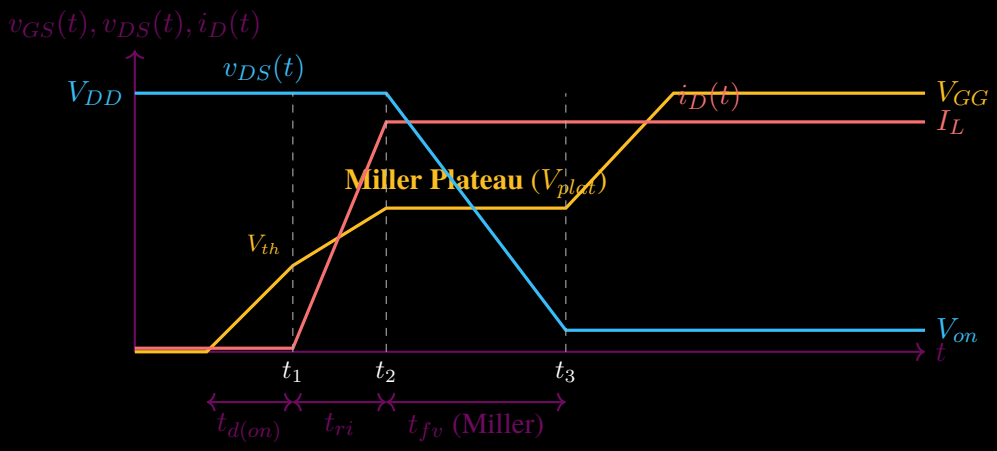


Figure 26: MOSFET turn-on switching dynamics illustrating the Miller Plateau during the drain voltage transition.

Physics of the Miller Plateau

During turn-on with a clamped inductive load:

1. **Interval $t_0 \to t_1$ (Turn-on Delay $t_{d(on)}$):** Gate current charges C_{gs} from 0 up to threshold voltage V_{th}. Drain current remains zero.
2. **Interval $t_1 \to t_2$ (Current Rise t_{ri}):** V_{GS} rises from V_{th} to the Miller plateau level $V_{plat} = V_{th} + I_L/g_m$. Drain current rises from 0 to full load current I_L.
3. **Interval $t_2 \to t_3$ (Voltage Fall t_{fv} — The Miller Plateau):** Once i_D reaches I_L, the freewheeling diode turns off, allowing v_{DS} to fall from V_{DD} toward zero. As v_{DS} plunges, the rapidly changing voltage across the drain-to-gate capacitance draws a displacement current:

$$i_{gd} = C_{gd} \cdot \left| \frac{dv_{DS}}{dt} \right| \tag{79}$$

   All gate driver current is diverted into discharging C_{gd}! Zero current is left to charge C_{gs}. Consequently, **the gate-to-source voltage V_{GS} remains completely flat** at the plateau voltage V_{plat} until v_{DS} finishes collapsing.
4. **Interval $t_3 \to t_4$:** V_{GS} resumes exponential charging toward the driver supply rail V_{GG}, driving the channel into deep ohmic conduction to minimize $R_{DS(on)}$.

Insulated Gate Bipolar Transistors (IGBTs)

The **IGBT** is a three-terminal, voltage-controlled minority-carrier (bipolar) power semiconductor switch. It was invented by B. Jayant Baliga to combine the **high input impedance and fast MOS gate control** of a Power MOSFET with the **massive current-carrying capacity and low forward voltage drop** of a bipolar transistor.

Physical Structure & Working Principle

The physical structure of an n-channel IGBT is identical to a vertical DMOSFET, with one revolutionary modification: **the n^+ drain substrate of the MOSFET is replaced by a heavily doped p^+ collector substrate!**

1. **Four-Layer Architecture:** Forms a $p^+ - n^- - p - n^+$ sandwich (resembling a thyristor, but controlled by a MOS gate).
2. **Conductivity Modulation in the Drift Layer:** When $V_{GE} > V_{GE(th)}$, the MOS gate induces an n-channel. Electrons flow from the n^+ emitter through the channel into the n^- drift region.
3. This electron current acts as the base current for the wide-base PNP transistor formed by p^+ (PNP emitter, i.e. the injector on the IGBT collector side) / n^- (drift region = PNP base) / p (body = PNP collector, tied to the IGBT emitter).
4. The p^+ substrate injects an immense swarm of **minority carrier holes** into the n^- drift region.
5. This floods the thick drift layer with an electron-hole plasma, causing massive **conductivity modulation**. The on-state voltage drop across the drift region collapses! A 1200 V, 100 A IGBT exhibits an on-state saturation voltage of merely:

$$V_{CE(sat)} \approx 1.5 - 2.0 \text{ V} \tag{80}$$

   compared to > 15 V for an equivalent silicon MOSFET!

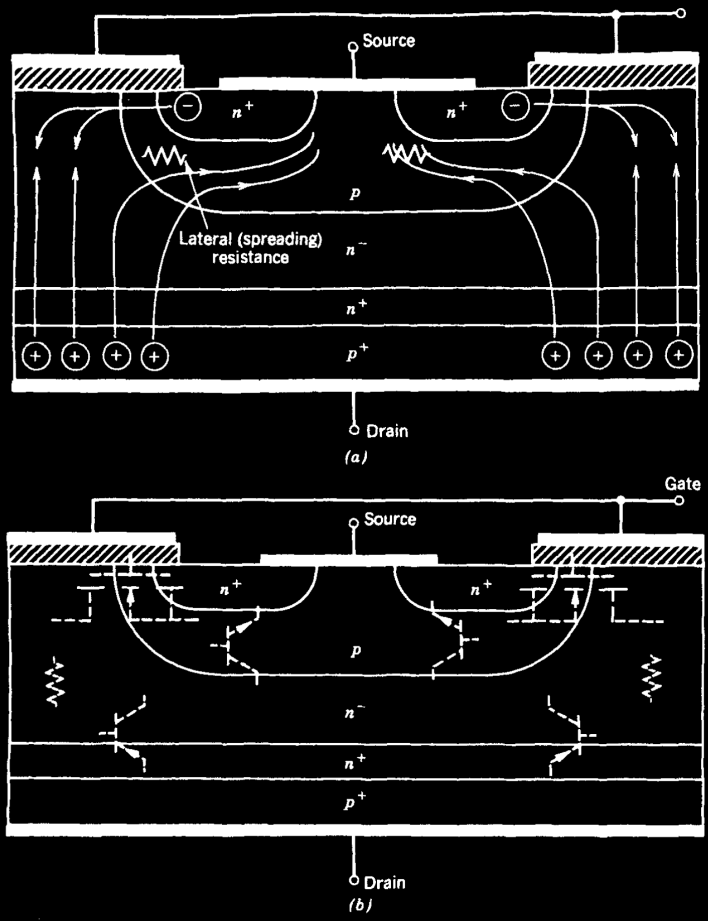


Figure 25-3 Vertical cross section of an IGBT showing (*a*) the on-state current flow paths and (*b*) the effective MOSFET and BJT operating portions of the structure.

Figure 27: Cross-sectional physical structure of an IGBT cell showing the p^+ collector injector substrate, n^+ buffer layer, n^- drift base, and the internal MOS gate channel alongside carrier flow. (Source: Ned Mohan, Fig. 25-3)

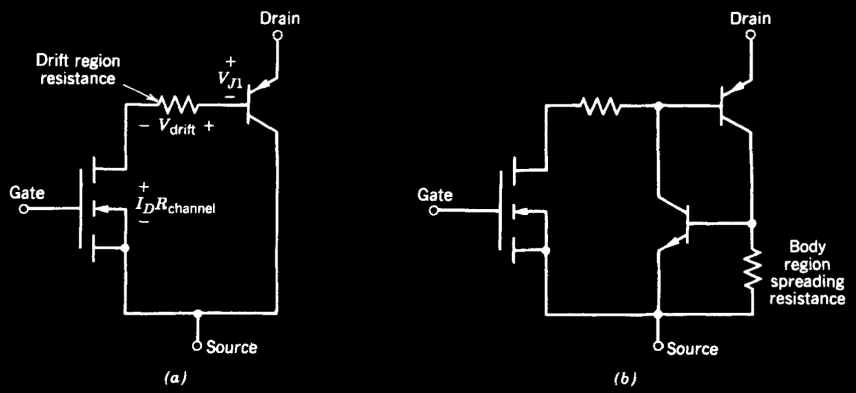


Figure 25-4 Equivalent circuits for the IGBT: (*a*) approximate equivalent circuit valid for normal operating conditions; (*b*) more complete IGBT equivalent circuit showing the

Figure 28: Equivalent circuit models of the IGBT: (a) approximate equivalent circuit valid under normal operation (MOSFET driving a wide-base PNP bipolar transistor); (b) complete equivalent circuit showing the internal parasitic NPN transistor and the thyristor latch-up loop across R_{body}. (Source: Ned Mohan, Fig. 25-4)

Equivalent Circuit & Parasitic Thyristor Latch-Up

The Danger of IGBT Latch-Up

The 4-layer structure contains a parasitic NPN transistor (Q_2) that pairs with Q_1 to form an **unwanted internal SCR**:

- The holes injected by Q_1 must travel laterally through the p-body to reach the emitter metallization.
- This lateral hole flow creates an ohmic voltage drop across the p-body resistance:

$$V_{BE2} = I_{\text{hole}} \cdot R_{\text{body}} \tag{81}$$

- If $V_{BE2} \geq 0.7$ V, the parasitic NPN transistor turns on, triggering **regenerative thyristor latch-up**!
- Once latched, the gate completely loses the ability to turn off collector current. The device suffers instantaneous short-circuit destruction.

Modern Mitigation: Modern Non-Punch-Through (NPT) and Field-Stop (FS) IGBTs employ heavily doped p^+ contact wells directly beneath the emitter to slash R_{body} to near zero, raising the dynamic latch-up threshold to several times rated surge current.

Dynamic Switching Characteristics & The Current Tail

While the IGBT turns ON quickly like a MOSFET, its **turn-off behavior is strictly limited by bipolar physics**.

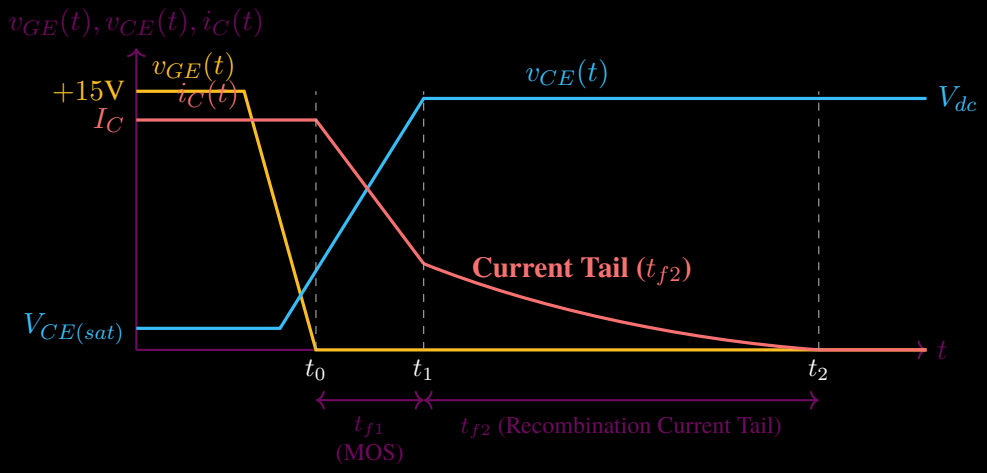


Figure 29: IGBT turn-off switching waveforms highlighting the persistent collector current tailing effect.

Physics of the Current Tail

When gate-to-emitter voltage v_{GE} is dropped below threshold:

1. **Rapid Current Drop (t_{f1}):** The MOS channel collapses instantaneously, extinguishing the electron current component ($I_{MOS} \approx 60\% - 70\%$ of total I_C).
2. **The Persistent Tail (t_{f2}):** With the MOS channel closed, the base of the wide-base PNP bipolar transistor is now **open-circuited**! The massive store of excess minority holes trapped in the thick n^- drift region has no external terminal through which to exit. These trapped carriers can disappear **only by natural recombination** ($\Delta p(t) = \Delta p_0 e^{-t/\tau_p}$).
3. **Thermal Impact:** Because v_{CE} has already risen to the full DC rail voltage V_{dc} while this tail current

persists, the instantaneous power dissipation $v_{CE}(t) \cdot i_C(t)$ during t_{f2} is massive!

4. **Frequency Limit:** The current tail drastically inflates the turn-off switching energy loss (E_{off}), restricting standard silicon IGBT switching frequencies to $\leq 20 - 30$ kHz.

Gate Drive Circuits for MOSFETs and IGBTs

Because power MOSFETs and IGBTs have insulated gates, their steady-state DC input resistance is $> 10^{10}\,\Omega$. However, dynamically, the gate represents a significant capacitive load ($C_{iss} \sim 1 - 10$ nF). To switch within tens of nanoseconds, the gate driver must source and sink high instantaneous peak currents:

$$I_{G,\text{peak}} = \frac{V_{GG} - V_{EE}}{R_G + R_{\text{int}}} \tag{82}$$

where R_G is the external series gate resistor and R_{int} is the internal polysilicon gate resistance.

Totem-Pole Push-Pull Driver

The fundamental discrete gate driver is the complementary BJT totem-pole emitter-follower circuit:

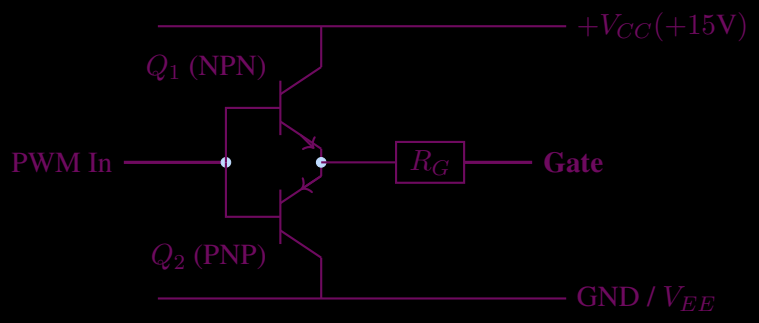


Figure 30: Totem-pole complementary push-pull gate driver providing high sourcing and sinking current capabilities.

- When PWM input is HIGH, Q_1 (NPN) conducts, rapidly charging the gate to $+V_{CC}$ with peak current limited only by R_G.
- When PWM input is LOW, Q_2 (PNP) conducts, rapidly dumping the gate charge to ground (V_{EE}).

Bootstrap Gate Driver Circuit (IR2110 Architecture)

In half-bridge and full-bridge inverters, the source/emitter of the **high-side switch** is tied to the switching phase node (V_S), which swings violently between 0 V and the high-voltage DC bus (V_{DC}). Driving the high-side switch requires a floating gate voltage referenced to V_S.

The **bootstrap circuit** generates this floating supply without an expensive isolated DC-DC converter.

Operating Principles of the Bootstrap Driver

1. **Charging Phase (Low-Side Switch ON):** When Q_L is turned ON, the phase node V_S is pulled to ground (0 V). Fast diode D_{boot} becomes forward-biased, charging bootstrap capacitor C_{boot} from V_{CC} (+15V) up to approximately $(V_{CC} - V_D)$.
2. **Floating Phase (High-Side Switch ON):** When Q_L turns OFF and Q_H turns ON, V_S leaps up to the DC bus voltage V_{DC}. Diode D_{boot} becomes reverse-biased, isolating the low-voltage supply. The capacitor C_{boot} floats with V_S, providing a stable $(V_S + 15\text{V})$ rail between pins V_B and V_S to keep Q_H fully enhanced.

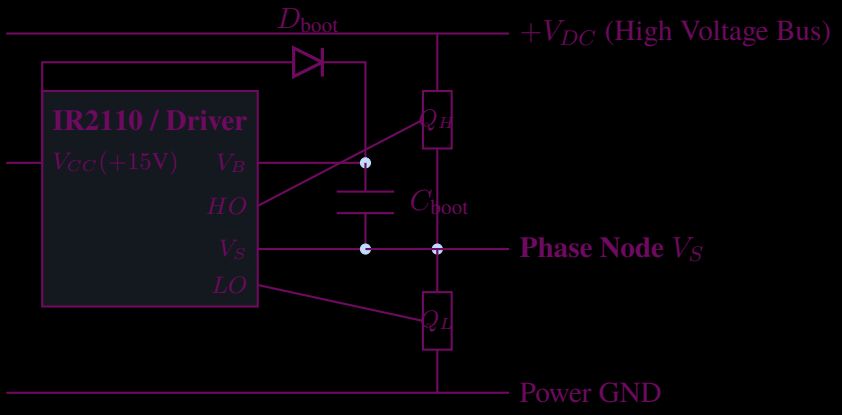


Figure 31: Bootstrap gate drive topology (IR2110 architecture) supplying the floating high-side gate charge.

3. **Duty Cycle Limitation:** Because C_{boot} can recharge **only when** Q_L **is ON** ($V_S = 0$), a bootstrap circuit **cannot operate at** 100% **duty cycle** ($D < 1$). Periodic low-side conduction is strictly required.

Bootstrap Capacitor Sizing Equation

The capacitor C_{boot} must supply the gate charge Q_g, the driver quiescent current I_{QBS}, level-shift charge Q_{LS}, and leakage currents during the maximum high-side on-time $t_{on,\text{max}}$, while limiting the voltage drop across C_{boot} to $\Delta V_{\text{boot}} \leq 0.5 - 1.0$ V:

$$C_{\text{boot}} \geq \frac{Q_g + I_{QBS} \cdot t_{on,\text{max}} + Q_{LS} + I_{\text{leak}} \cdot t_{on,\text{max}}}{\Delta V_{\text{boot}}} \tag{83}$$

A typical design rule of thumb selects $C_{\text{boot}} \geq 10$ to $15 \times C_{iss}$.

Modern Power Devices: Wide Bandgap (WBG) Semiconductors

Physical Limitations of Silicon

For over six decades, Silicon (Si) has served as the bedrock of power semiconductor technology. However, silicon is rapidly approaching its fundamental, immutable physical limits imposed by its material properties:

1. **Narrow Bandgap ($E_g = 1.12$ eV):** Under high temperatures ($T_j > 150$°C), thermal energy excites valence electrons across the narrow bandgap into the conduction band ($n_i \propto T^{3/2} e^{-E_g/2kT}$), causing excessive intrinsic leakage currents and thermal runaway.

2. **Low Critical Breakdown Electric Field ($E_c \approx 0.3$ MV/cm):** Silicon can withstand an electric field of only 30 V/μm before avalanche impact ionization occurs. To block thousands of volts, the n^- drift region must be made extraordinarily thick and lightly doped ($W_d \approx 2V_{BR}/E_c \propto V_{BR}$ and $N_d \approx \epsilon_s E_c^2/(2qV_{BR}) \propto 1/V_{BR}$).

3. **The Silicon Unipolar Limit:** According to Baliga's Figure of Merit (BFOM), the minimum theoretical specific on-resistance of a unipolar drift region is:

$$R_{on,sp} = \frac{4V_{BR}^2}{\epsilon_s \cdot \mu_n \cdot E_c^3} \tag{84}$$

Because E_c is low for silicon, $R_{on,sp}$ scales aggressively with V_{BR}^2 (with E_c held constant; real silicon follows about $V_{BR}^{2.5}$, see the MOSFET section). For ratings above 600 V, silicon MOSFETs become prohibitively resistive, forcing the use of bipolar devices (IGBTs) with their inherent switching frequency limitations (current tailing).

4. **Moderate Thermal Conductivity ($k = 1.5$ W/cm · K):** Silicon cannot evacuate heat rapidly, requiring large, heavy heat sinks and forced liquid cooling systems.

Wide Bandgap Semiconductors: SiC and GaN

Wide Bandgap (WBG) semiconductors—principally **Silicon Carbide (4H-SiC)** and **Gallium Nitride (GaN)**—possess electronic bandgaps approximately three times wider than silicon.

Table 6: Fundamental Material Properties: Silicon (Si) vs. 4H-SiC vs. Gallium Nitride (GaN)

Physical Property	Silicon (Si)	Silicon Carbide (4H-SiC)	Gallium Nitride (GaN)
Bandgap Energy (E_g)	1.12 eV	3.26 eV (2.9×)	3.40 eV (3.0×)
Critical Breakdown Field (E_c)	0.3 MV/cm	2.8 MV/cm (9.3×)	3.3 MV/cm (11.0×)
Electron Mobility (μ_n)	1400 $\text{cm}^2/\text{V} \cdot \text{s}$	900 $\text{cm}^2/\text{V} \cdot \text{s}$	$1500 - 2000$ $\text{cm}^2/\text{V} \cdot \text{s}$ (2DEG)
Saturated Electron Velocity (v_{sat})	1.0×10^7 cm/s	2.0×10^7 cm/s (2.0×)	2.5×10^7 cm/s (2.5×)
Thermal Conductivity (k)	1.5 W/cm · K	4.9 W/cm · K (3.3×)	$1.5 - 2.0$ W/cm · K
Dielectric Constant (ϵ_r)	11.8	9.7	8.9
Max Operating Temp ($T_{j,\text{max}}$)	150°C	> 200°C $- 250$°C	> 200°C
Baliga's FOM ($\propto \epsilon\mu E_c^3$)	1.0 (Baseline)	≈ 500	$\approx 1000 - 1500$

The Revolutionary Impact of a $10\times$ Higher Breakdown Field (E_c)

The breakdown electric field E_c enters the unipolar specific on-resistance equation to the **third power** (E_c^3):

$$R_{on,sp} \propto \frac{1}{E_c^3} \tag{85}$$

Because SiC and GaN exhibit E_c values roughly $10\times$ greater than silicon:

1. $10\times$ **Thinner Drift Layer** ($W_d \propto V_{BR}/E_c$)**:** A 1200 V SiC drift layer requires only $\sim 10\,\mu$m of thickness, compared to $\sim 100\,\mu$m in silicon.
2. $100\times$ **Higher Doping Concentration** ($N_d \propto E_c^2/V_{BR}$)**:** The drift layer can be doped 100 times more heavily without premature breakdown.
3. **Drastic Resistance Reduction:** Combining a $10\times$ thinner layer with $100\times$ higher conductivity reduces the theoretical drift resistance by up to $1000\times$ compared to a silicon MOSFET!
4. **Unipolar Operation at High Voltage:** High-voltage SiC MOSFETs ($1200\text{ V} - 3300\text{ V}$) can operate as pure unipolar majority-carrier switches, completely eliminating the minority carrier current tailing of IGBTs!

Silicon Carbide (SiC) MOSFETs

Silicon Carbide (4H-SiC) is the premier material for high-power, high-voltage ($650\text{ V} - 3300\text{ V}+$) conversion:

- **Direct Replacement of Silicon IGBTs:** SiC MOSFETs replace 1200 V IGBTs in EV traction inverters, slashing total inverter power losses by $60\% - 75\%$ and extending EV driving range by $5\% - 10\%$ on the same battery capacity.
- **Superb Thermal Conductivity** ($k = 4.9$ **W/cm·K):** SiC conducts heat three times better than silicon, enabling dramatic miniaturization of heat sinks.

- **Extremely Low Switching Losses:** Because SiC MOSFETs are unipolar, E_{off} is dictated purely by capacitance discharge without any carrier recombination tail. Operating frequencies can be pushed from 15 kHz (IGBT) to 100 kHz $-$ 300 kHz.
- **Body Diode Nuance:** The wide bandgap gives the intrinsic SiC body diode a high forward knee voltage ($V_F \approx 3.2 - 4.0$ V). If current freewheels through the body diode during dead-time, conduction losses are high. Engineers solve this using **synchronous rectification** (turning on the SiC channel during reverse freewheeling) or adding an anti-parallel SiC Schottky barrier diode.
- **Gate Drive Levels:** Because SiC has a lower channel inversion layer mobility due to SiO_2/SiC interface traps, a higher gate-to-source voltage ($+18$ V to $+20$ V) is required to achieve minimal $R_{DS(on)}$. A negative turn-off bias (-4 V to -5 V) is universally employed to prevent spurious Miller turn-on under extreme dv/dt ($>$ 50 V/ns).

Gallium Nitride (GaN) HEMTs

Gallium Nitride power devices are constructed as lateral **High Electron Mobility Transistors (HEMTs)**.

Physics of the Two-Dimensional Electron Gas (2DEG)

Unlike silicon MOSFETs that require gate voltage to invert the channel, the GaN HEMT features a spontaneous, ultra-conductive channel known as the **Two-Dimensional Electron Gas (2DEG)**:

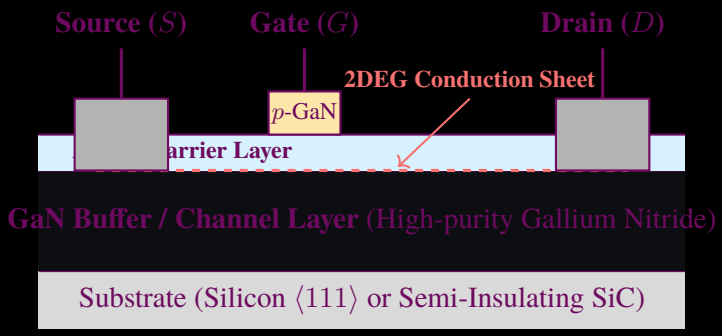


Figure 32: Heterojunction architecture of a lateral GaN High Electron Mobility Transistor (HEMT) showing the spontaneously formed 2DEG conduction sheet.

1. **Heterojunction Formation:** When a thin layer of Aluminum Gallium Nitride (AlGaN) is grown epitaxially on high-purity GaN, the difference in spontaneous polarization and the massive piezoelectric stress at the lattice-mismatched interface induce an intense internal electric field.
2. **Carrier Confinement:** This field bends the conduction band below the Fermi level at the interface, trapping electrons in a quantum well just a few nanometers thick: the **2DEG**.
3. **Phenomenal Mobility Without Doping:** Unlike silicon or SiC where high conductivity requires intentional donor doping (which causes ionized impurity scattering), the 2DEG is populated purely by polarization charges! Because the channel is free of dopant atoms, electrons glide through the crystal lattice with virtually zero scattering, achieving an extraordinary electron mobility:

$$\mu_{2DEG} \approx 1500 - 2000 \text{ cm}^2/\text{V} \cdot \text{s} \tag{86}$$

4. **Depletion-Mode (d-mode) vs. Enhancement-Mode (e-mode):**
   - **d-mode (Normally-ON):** Because the 2DEG is formed naturally by crystal physics, a standard GaN HEMT conducts heavy current at $V_{GS} = 0$ V. In power electronics, normally-on switches present severe safety hazards during power-up.

- **p-GaN Gate e-mode (Normally-OFF):** Growing a heavily doped p-type GaN layer beneath the gate lifts the conduction band above the Fermi level, completely depleting the 2DEG directly under the gate at zero bias ($V_{th} \approx +1.4 - 1.7$ V). Applying $V_{GS} \geq 5$ V re-establishes the 2DEG channel.

The GaN Cascode Configuration

Another widely adopted approach to achieve safe, normally-off operation is the **Cascode Configuration**:

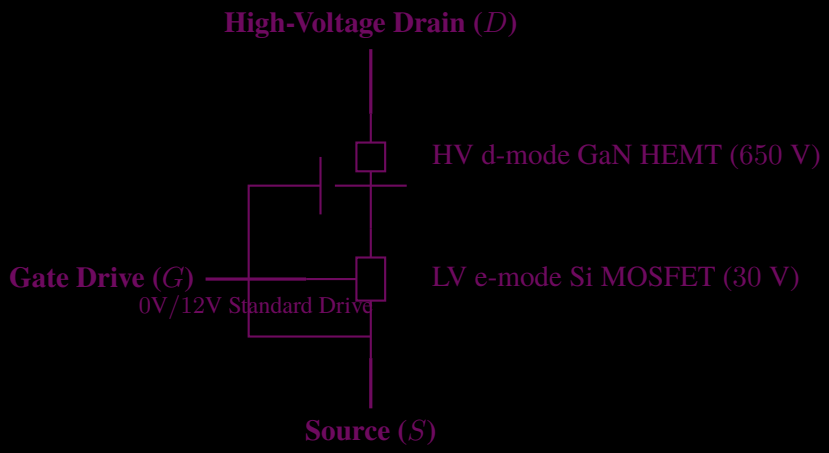


Figure 33: GaN Cascode configuration pairing a low-voltage Silicon MOSFET with a high-voltage depletion-mode GaN HEMT.

Operating Principle of the GaN Cascode

1. When Gate Drive $V_G = 0$ V, the low-voltage Si MOSFET is OFF. The load current charges the capacitance at the node between the two devices (the GaN source), so this node voltage starts to rise.
2. The GaN gate is tied to ground (the Si MOSFET source), so $V_{GS,\text{GaN}} = -v_{node}$. As soon as the node reaches the GaN pinch-off magnitude $|V_p|$ (typically of the order of 10 V; see the datasheet), $V_{GS,\text{GaN}} \leq V_p$ and the HEMT turns OFF. The node therefore settles near $|V_p|$, which is why a 30 V Si MOSFET is enough, while the GaN device blocks the remaining ≈ 640 V of the 650 V bus.
3. When $V_G = 12$ V, the Si MOSFET turns ON hard, pulling the GaN source to ground ($V_{GS,\text{GaN}} \to 0$ V). The GaN HEMT immediately turns ON.
4. **Huge Engineering Benefit:** The cascode uses standard, rugged Silicon gate driver ICs ($0-12$ V) and has high noise immunity, while delivering the ultra-fast switching speed of GaN. (Reverse-recovery charge is *small but not zero*: it comes from the low-voltage Si MOSFET body diode, and is still far below that of a 650 V Si MOSFET.)

Key Features & Advantages of GaN HEMTs

- **Zero Reverse Recovery ($Q_{rr} \approx 0$) in e-mode devices:** Because the 2DEG channel has **zero minority carrier storage**, there is no reverse recovery delay when reverse current ceases. The device freewheels in the third quadrant by channel conduction without reverse recovery loss.
- **Third-quadrant voltage drop:** There is no body diode. Reverse conduction needs the gate-to-drain voltage to exceed V_{th}, so the reverse drop is about $V_{th} + |V_{GS,off}|$ (typically $\approx 2-3$ V at 0 V off-bias, more with a negative off-bias). Keep dead-time short to limit this loss.
- **Extreme Switching Frequencies (500 kHz $-$ 10 MHz):** The input capacitance (C_{iss}) and output capacitance (C_{oss}) of a GaN HEMT are an order of magnitude smaller than a silicon MOSFET of identical current rating. This allows switching frequencies an order of magnitude higher with minimal power loss.

- **Radical Miniaturization of Magnetics:** Operating at megahertz frequencies shrinks the physical volume of inductors and transformers by 80%, enabling the pocket-sized USB-PD 65W–240W laptop chargers that have revolutionized consumer electronics.

Solved Numerical Examination Problems

This section presents nine fully worked, textbook-grade numerical problems spanning every quantitative concept in Module 1. Each problem provides the full problem statement, systematic identification of given data, rigorous step-by-step arithmetic, explicit unit conversions, and key engineering takeaways.

Problem 1: Thermal Modeling and Heat Sink Sizing

Problem 1: Thermal Resistance Network & Heat Sink Selection

A power MOSFET packaged in a standard TO-220 case operates in a DC-DC switching regulator and dissipates an average continuous power of $P_D = 28$ W. The maximum allowable junction temperature specified by the manufacturer is $T_{j,\max} = 150°\text{C}$, and the ambient air temperature inside the converter enclosure is $T_a = 45°\text{C}$.
The device datasheet provides the junction-to-case thermal resistance as $R_{\theta,jc} = 1.25°\text{C/W}$. The transistor is mechanically clamped to an extruded aluminum heat sink using a thermal silicone pad and zinc oxide grease, yielding an interface case-to-sink thermal resistance of $R_{\theta,cs} = 0.35°\text{C/W}$.
Determine:

(a) The maximum permissible thermal resistance of the heat sink to ambient ($R_{\theta,sa}$).

(b) The operating case temperature (T_c) and heat sink surface temperature (T_s) at rated load.

(c) The maximum power the MOSFET can safely dissipate if operated **without a heat sink** in still air, given that the case-to-ambient thermal resistance is $R_{\theta,ca} = 55°\text{C/W}$.

Solution:
Step 1: Identify Given Data

- Average power dissipation: $P_D = 28$ W
- Maximum permissible junction temperature: $T_{j,\max} = 150°\text{C}$
- Ambient temperature: $T_a = 45°\text{C}$
- Junction-to-case thermal resistance: $R_{\theta,jc} = 1.25°\text{C/W}$
- Case-to-sink thermal resistance: $R_{\theta,cs} = 0.35°\text{C/W}$
- Free-air case-to-ambient resistance: $R_{\theta,ca} = 55°\text{C/W}$

Step 2: Calculate Maximum Allowable Total Thermal Resistance ($R_{\theta,ja}$) Applying the steady-state thermal Ohm's law:

$$T_j - T_a = P_D \cdot R_{\theta,ja} \implies R_{\theta,ja} = \frac{T_{j,\max} - T_a}{P_D} \tag{87}$$

Substituting numerical values:

$$R_{\theta,ja} = \frac{150°\text{C} - 45°\text{C}}{28\ \text{W}} = \frac{105°\text{C}}{28\ \text{W}} = 3.75°\text{C/W} \tag{88}$$

Step 3: Calculate Required Heat Sink Thermal Resistance ($R_{\theta,sa}$) The total series thermal resistance is:

$$R_{\theta,ja} = R_{\theta,jc} + R_{\theta,cs} + R_{\theta,sa} \tag{89}$$

Rearranging to solve for $R_{\theta,sa}$:

$$R_{\theta,sa} \leq R_{\theta,ja} - (R_{\theta,jc} + R_{\theta,cs}) \tag{90}$$

$$R_{\theta,sa} \leq 3.75 - (1.25 + 0.35) = 3.75 - 1.60 = \mathbf{2.15}^\circ\text{C/W} \tag{91}$$

Takeaway: The design engineer must select an extruded aluminum heat sink with a thermal resistance rating of 2.15°**C/W or lower**.

Step 4: Calculate Operating Case and Sink Temperatures The case temperature is:

$$T_c = T_j - P_D \cdot R_{\theta,jc} = 150^\circ\text{C} - (28\ \text{W} \times 1.25^\circ\text{C/W}) = 150 - 35 = \mathbf{115.0}^\circ\text{C} \tag{92}$$

The heat sink temperature is:

$$T_s = T_c - P_D \cdot R_{\theta,cs} = 115.0^\circ\text{C} - (28\ \text{W} \times 0.35^\circ\text{C/W}) = 115.0 - 9.8 = \mathbf{105.2}^\circ\text{C} \tag{93}$$

Checking against ambient:

$$T_s = T_a + P_D \cdot R_{\theta,sa} = 45 + (28 \times 2.15) = 45 + 60.2 = 105.2^\circ\text{C} \quad \text{(Confirmed)} \tag{94}$$

Step 5: Maximum Power Dissipation Without Heat Sink In free air without a heat sink:

$$R_{\theta,ja(\text{free air})} = R_{\theta,jc} + R_{\theta,ca} = 1.25 + 55 = 56.25^\circ\text{C/W} \tag{95}$$

The maximum safe continuous power is:

$$P_{D,\text{max(no sink)}} = \frac{T_{j,\text{max}} - T_a}{R_{\theta,ja(\text{free air})}} = \frac{150 - 45}{56.25} = \frac{105}{56.25} = \mathbf{1.867}\ \text{W} \tag{96}$$

Critical Insight: Without a heat sink, dissipating just 1.87 W heats the silicon junction to its absolute destruction limit (150°C). Sizing the heat sink correctly increases power capability by fifteen-fold!

Problem 2: Power Diode Reverse Recovery Dynamics

Problem 2: Reverse Recovery Transient Calculations

A power diode (a medium-speed device, slower than a true fast-recovery class) conducts a forward DC current of $I_F = 80$ A. During converter turn-off, the circuit applies a reverse voltage $V_R = 450$ V, forcing the diode current to ramp down at a constant rate of $di/dt = 120\ \text{A}/\mu\text{s} = 1.2 \times 10^8$ A/s.
The diode exhibits a total reverse recovery time of $t_{rr} = 2.5\,\mu$s with a softness factor of $S = t_b/t_a = 0.25$.
Calculate:

(a) The storage interval (t_a) and fall/decay interval (t_b).

(b) The peak reverse recovery current (I_{RR}).

(c) The total reverse recovery stored charge (Q_{RR}).

(d) The reverse recovery power loss dissipated in the diode if operated at switching frequency $f_{sw} = 20$ kHz.

Solution:
Step 1: Calculate t_a and t_b We are given:

$$t_{rr} = t_a + t_b = 2.5\,\mu\text{s}, \qquad S = \frac{t_b}{t_a} = 0.25 \implies t_b = 0.25 t_a \tag{97}$$

Substituting t_b into the total time equation:

$$t_a + 0.25t_a = 1.25t_a = 2.5\,\mu\text{s} \tag{98}$$

$$t_a = \frac{2.5\,\mu\text{s}}{1.25} = \mathbf{2.0}\,\mu\text{s} \tag{99}$$

$$t_b = 0.25 \times 2.0\,\mu\text{s} = \mathbf{0.5}\,\mu\text{s} \tag{100}$$

Step 2: Calculate Peak Reverse Recovery Current (I_{RR}) During the storage interval t_a, current ramps from zero to $-I_{RR}$ at rate di/dt:

$$I_{RR} = \left(\frac{di}{dt}\right) \cdot t_a \tag{101}$$

$$I_{RR} = (120\ \text{A}/\mu\text{s}) \times (2.0\,\mu\text{s}) = \mathbf{240.0}\ \text{A} \tag{102}$$

Observation: The peak reverse recovery current (240 A) is three times larger than the original forward current (80 A)!

Step 3: Calculate Stored Reverse Recovery Charge (Q_{RR}) Using the triangular approximation:

$$Q_{RR} = \frac{1}{2} \cdot I_{RR} \cdot t_{rr} \tag{103}$$

$$Q_{RR} = \frac{1}{2} \times 240\ \text{A} \times (2.5 \times 10^{-6}\ \text{s}) = \mathbf{300.0}\,\mu\text{C} = 0.30\ \text{mC} \tag{104}$$

Step 4: Calculate Reverse Recovery Energy and Power Loss Power dissipation occurs during the fall interval t_b while voltage rises across the recovering junction:

$$E_{rr} \approx \frac{1}{6} V_R \cdot I_{RR} \cdot t_b = \frac{1}{6} \times 450\ \text{V} \times 240\ \text{A} \times (0.5 \times 10^{-6}\ \text{s}) \tag{105}$$

$$E_{rr} = \frac{1}{6} \times 108000 \times 0.5 \times 10^{-6} = 18000 \times 0.5 \times 10^{-6} = \mathbf{0.009}\ \text{J} = 9.0\ \text{mJ} \tag{106}$$

Multiplying by switching frequency $f_{sw} = 20$ kHz:

$$P_{rr} = E_{rr} \cdot f_{sw} = (9.0 \times 10^{-3}\ \text{J}) \times (20 \times 10^{3}\ \text{Hz}) = \mathbf{180.0}\ \text{W} \tag{107}$$

Problem 3: SCR Two-Transistor Analogy and Anode Current

Problem 3: Regenerative Current Gain and Latch-Up Mechanics

In a two-transistor equivalent model of an SCR, the common-base current gains of the PNP transistor (Q_1) and NPN transistor (Q_2) under low-current blocking conditions are $\alpha_1 = 0.40$ and $\alpha_2 = 0.45$, respectively. The collector-base reverse saturation leakage currents are $I_{CBO1} = 20\,\mu$A and $I_{CBO2} = 30\,\mu$A.

Determine:

(a) The off-state forward leakage anode current (I_A) when gate current $I_G = 0$.

(b) The anode current if a gate current pulse of $I_G = 30$ mA is injected, assuming this pulse causes α_2 to increase to 0.52 while α_1 remains at 0.40.

(c) If the NPN current gain depends on gate current according to $\alpha_2 = 0.45 + 0.003 \cdot I_G$ (where I_G is in mA) and α_1 remains constant at 0.40, calculate the critical gate trigger current (I_{GT}) required to initiate regenerative latch-up ($\alpha_1 + \alpha_2 \to 1$).

Solution:
Step 1: Anode Current with Zero Gate Signal ($I_G = 0$) The master two-transistor anode current equation is:

$$I_A = \frac{\alpha_2 I_G + I_{CBO1} + I_{CBO2}}{1 - (\alpha_1 + \alpha_2)} \tag{108}$$

Substituting $I_G = 0$, $\alpha_1 = 0.40$, $\alpha_2 = 0.45$, and leakage currents:

$$I_A = \frac{0 + 20\,\mu\text{A} + 30\,\mu\text{A}}{1 - (0.40 + 0.45)} = \frac{50\,\mu\text{A}}{1 - 0.85} = \frac{50\,\mu\text{A}}{0.15} = \mathbf{333.33}\,\mu\text{A} = 0.333\text{ mA} \tag{109}$$

Physical Meaning: In the forward blocking state, the total leakage current is amplified by the regenerative loop factor $\frac{1}{1-(\alpha_1+\alpha_2)} = 6.67$, but remains sub-milliampere.
Step 2: Anode Current with Injected Gate Pulse ($I_G = 30$ mA) Here, $\alpha_1 = 0.40$ and $\alpha_2 = 0.52$, giving:

$$\alpha_1 + \alpha_2 = 0.40 + 0.52 = 0.92 \tag{110}$$

Numerator:

$$\alpha_2 I_G + I_{CBO1} + I_{CBO2} = (0.52 \times 30\text{ mA}) + 0.05\text{ mA} = 15.60\text{ mA} + 0.05\text{ mA} = 15.65\text{ mA} \tag{111}$$

Denominator:

$$1 - (\alpha_1 + \alpha_2) = 1 - 0.92 = 0.08 \tag{112}$$

Therefore:

$$I_A = \frac{15.65\text{ mA}}{0.08} = \mathbf{195.63}\text{ mA} \tag{113}$$

Step 3: Critical Gate Current for Latch-Up Regenerative latch-up occurs when the loop gain reaches unity:

$$\alpha_1 + \alpha_2 = 1.0 \tag{114}$$

Substituting $\alpha_1 = 0.40$ and $\alpha_2(I_G) = 0.45 + 0.003 \cdot I_G$:

$$0.40 + (0.45 + 0.003 \cdot I_G) = 1.0 \tag{115}$$

$$0.85 + 0.003 \cdot I_G = 1.0 \implies 0.003 \cdot I_G = 0.15 \tag{116}$$

$$I_G = \frac{0.15}{0.003} = \mathbf{50.0}\text{ mA} \tag{117}$$

Conclusion: When gate current reaches 50 **mA**, the sum of the gains reaches unity ($\alpha_1 + \alpha_2 = 1$), causing the denominator to vanish ($I_A \to \infty$ limited only by load resistance). The SCR latches ON.

Problem 4: SCR Minimum Gate Pulse Duration with Inductive Load

Problem 4: Gate Pulse Width Requirement for Inductive Loads

An SCR is used to switch a 230 V DC supply into an inductive load consisting of resistance $R = 12\,\Omega$ and inductance $L = 0.24$ H. The latching current of the SCR is $I_L = 80$ mA.
Determine:

(a) The minimum gate pulse duration ($t_{p,\text{min}}$) required to ensure reliable latch-up of the SCR.

(b) If the circuit inductance is increased to $L = 0.96$ H due to a change in motor winding configuration, calculate the new required gate pulse width.

(c) If the trigger circuit provides a standard pulse width of only $50\,\mu$s, determine whether the SCR will

turn on in case (a) and case (b).

Solution:

Step 1: Circuit Differential Equation When the gate trigger pulse arrives and the SCR starts conducting, the circuit is governed by:

$$L\frac{di_A}{dt} + R \cdot i_A = V_s \tag{118}$$

Solving with initial condition $i_A(0) = 0$:

$$i_A(t) = \frac{V_s}{R}\left(1 - e^{-Rt/L}\right) \tag{119}$$

For reliable latch-up, the gate pulse must remain active until the anode current reaches the latching current:

$$i_A(t_{p,\text{min}}) = I_L = 80 \text{ mA} = 0.08 \text{ A} \tag{120}$$

Step 2: Exact Calculation of $t_{p,\text{min}}$ for Case (a) Given $V_s = 230$ V, $R = 12\,\Omega$, $L = 0.24$ H:

$$\frac{V_s}{R} = \frac{230}{12} = 19.167 \text{ A} \qquad \text{(Steady-state current)} \tag{121}$$

The time constant is:

$$\tau = \frac{L}{R} = \frac{0.24}{12} = 0.02 \text{ s} = 20 \text{ ms} \tag{122}$$

Setting $i_A(t) = I_L$:

$$0.08 = 19.167\left(1 - e^{-t/0.02}\right) \implies 1 - e^{-t/0.02} = \frac{0.08}{19.167} = 0.004174 \tag{123}$$

$$e^{-t/0.02} = 1 - 0.004174 = 0.995826 \tag{124}$$

Taking the natural logarithm:

$$-\frac{t}{0.02} = \ln(0.995826) = -0.004183 \implies t_{p,\text{min}} = 0.02 \times 0.004183 = \mathbf{83.66}\,\mu\text{s} \tag{125}$$

Linear Approximation Check: Because $t \ll \tau$, we can use the linear slope $i_A(t) \approx \frac{V_s}{L} \cdot t$:

$$t_{p,\text{min}} \approx \frac{L \cdot I_L}{V_s} = \frac{0.24 \times 0.08}{230} = \frac{0.0192}{230} = 83.48\,\mu\text{s} \quad (\text{Error } < 0.2\%) \tag{126}$$

Step 3: Minimum Pulse Width for Case (b) ($L = 0.96$ H) Using the accurate linear relation:

$$t_{p,\text{min}} \approx \frac{L \cdot I_L}{V_s} = \frac{0.96 \times 0.08}{230} = \frac{0.0768}{230} = \mathbf{333.91}\,\mu\text{s} \tag{127}$$

Step 4: Verification Against $50\,\mu$s Trigger Pulse

- **Case (a) ($L = 0.24$ H):** Required width is $83.66\,\mu$s. Because $50\,\mu\text{s} < 83.66\,\mu\text{s}$, the anode current reaches only:

$$i_A(50\,\mu\text{s}) \approx \frac{230 \times 50 \times 10^{-6}}{0.24} = 47.9 \text{ mA} < I_L(80 \text{ mA}) \tag{128}$$

  The SCR will FAIL to turn on!

- **Case (b) ($L = 0.96$ H):** Required width is $333.9\,\mu$s. The current reaches only 11.98 mA $\ll 80$ mA. **The SCR will FAIL to turn on!**

Design Fix: Add a parallel bleeder resistor across the load or generate a pulse train of duration $> 350\,\mu$s.

Problem 5: SCR Snubber Circuit Design (di/dt and dv/dt)

Problem 5: Complete Snubber Network Design

An SCR operates in a single-phase bridge converter connected to a $V_s = 230$ V, 50 Hz AC mains supply. The maximum peak voltage including line surges is $V_m = \sqrt{2} \times 230 \times 1.2 = 390$ V.
The device ratings are:

- Maximum rate of rise of current: $(di/dt)_{\text{max}} = 40\ \text{A}/\mu\text{s} = 4.0 \times 10^7\ \text{A/s}$
- Maximum rate of rise of forward voltage: $(dv/dt)_{\text{max}} = 120\ \text{V}/\mu\text{s} = 1.2 \times 10^8\ \text{V/s}$
- Maximum peak discharge current into the snubber: $I_{T,\text{max}} = 60$ A

Design the snubber protection network:

(a) Sizing of series inductor L_s for di/dt protection, and check whether it is also adequate for the dv/dt and discharge-current limits.

(b) Selection of the snubber resistor R_s from the discharge-current and dv/dt limits.

(c) Sizing of the snubber capacitor C_s for optimal damping ($\xi = 0.65$).

(d) Verify all ratings and calculate the power dissipated in the snubber at 50 Hz.

Solution:
Step 1: Minimum series inductance for di/dt. At turn-on the inductor alone limits the current slope to V_m/L_s:

$$L_s \geq \frac{V_m}{(di/dt)_{\text{max}}} = \frac{390\ \text{V}}{4.0 \times 10^7\ \text{A/s}} = \mathbf{9.75}\,\mu\text{H} \tag{129}$$

Step 2: Bounds on R_s. The discharge current at firing is about V_m/R_s, so

$$R_s \geq \frac{V_m}{I_{T,\text{max}}} = \frac{390}{60} = 6.50\,\Omega \tag{130}$$

The dv/dt limit requires $R_s \leq L_s(dv/dt)_{\text{max}}/V_m$. With $R_s \geq 6.50\,\Omega$ this demands

$$L_s \geq \frac{R_s V_m}{(dv/dt)_{\text{max}}} = \frac{6.50 \times 390}{1.2 \times 10^8} = 21.1\,\mu\text{H} \tag{131}$$

so $L_s = 9.75\,\mu$H from Step 1 is *not enough*; the dv/dt and discharge-current limits are the binding ones. Choose the standard value $\mathbf{L_s = 25\,\mu H}$.
Step 3: Choose R_s. With $L_s = 25\,\mu$H:

$$6.50\,\Omega \leq R_s \leq \frac{25 \times 10^{-6} \times 1.2 \times 10^8}{390} = 7.69\,\Omega \quad \Longrightarrow \quad R_s = \mathbf{6.8}\,\Omega \text{ (standard value)} \tag{132}$$

Step 4: Snubber capacitor for $\xi = 0.65$.

$$C_s = L_s\left(\frac{2\xi}{R_s}\right)^2 = 25 \times 10^{-6} \times \left(\frac{1.3}{6.8}\right)^2 = 25 \times 10^{-6} \times 0.03655 = 0.914\,\mu\text{F} \tag{133}$$

Select a standard $\mathbf{C_s = 1.0\,\mu F}$ polypropylene pulse capacitor; this gives $\xi = \frac{6.8}{2}\sqrt{1.0/25} = 0.68$.
Step 5: Verification of all ratings.

- di/dt: $V_m/L_s = 390/25\mu\text{H} = 15.6\ \text{A}/\mu\text{s} < 40\ \text{A}/\mu\text{s}$ ✓
- dv/dt: $V_m R_s/L_s = 390 \times 6.8/25\mu\text{H} = 106\ \text{V}/\mu\text{s} < 120\ \text{V}/\mu\text{s}$ ✓

- Discharge current: $V_m/R_s = 390/6.8 = 57.4\text{ A} < 60\text{ A}$ ✓

Step 6: Snubber power dissipation (conservative).

$$P_{\text{snub}} \approx C_s V_m^2 f = (1.0 \times 10^{-6}) \times (390)^2 \times 50 = \mathbf{7.6}\text{ W} \tag{134}$$

(Only the firing discharge, $\frac{1}{2}C_s V_m^2 f = 3.8$ W, is a certain loss, so a 10 W wirewound non-inductive resistor is comfortable.)

Pitfall: why not $L_s = 10\,\mu$H, $R_s = 6.8\,\Omega$?

With $L_s = 10\,\mu$H the initial slope is $V_m R_s/L_s = 390 \times 6.8/10\mu\text{H} = 265\text{ V}/\mu\text{s}$, more than twice the $120\text{ V}/\mu\text{s}$ rating, even though R_s satisfies the discharge-current limit. Always check the dv/dt bound $R_s \le L_s(dv/dt)_{\max}/V_m$ after choosing R_s from the current limit.

Problem 6: UJT Relaxation Oscillator Triggering Circuit Design

Problem 6: Complete UJT Firing Circuit Design

A Uni-Junction Transistor (UJT) is employed in a relaxation oscillator circuit to trigger the SCRs of a phase-controlled converter. The circuit is powered by a stable DC rail voltage of $V_{BB} = 20$ V.
The UJT parameters from the manufacturer datasheet are:

- Intrinsic standoff ratio: $\eta = 0.65$
- Interbase resistance: $R_{BB} = 7.5\text{ k}\Omega$
- Emitter diode forward voltage drop: $V_D = 0.7$ V
- Peak point emitter current: $I_P = 10\,\mu$A
- Valley point emitter current: $I_V = 8$ mA
- Valley point voltage: $V_V = 1.5$ V

Calculate:

(a) The peak point voltage V_P.

(b) The allowable range of emitter resistance ($R_{E,\text{min}} \le R_E \le R_{E,\text{max}}$) to ensure sustained relaxation oscillation.

(c) If an emitter charging resistor of $R_E = 33\text{ k}\Omega$ is chosen, calculate the timing capacitance C_E needed to produce a trigger frequency of $f = 500$ Hz.

Solution:
Step 1: Calculate Peak Point Voltage (V_P)

$$V_P = \eta V_{BB} + V_D = (0.65 \times 20\text{ V}) + 0.7\text{ V} = 13.0 + 0.7 = \mathbf{13.7}\text{ V} \tag{135}$$

Step 2: Determine Stable Range of R_E To guarantee that the load line intersects the negative resistance region of the UJT emitter characteristic:

- **Upper Bound ($R_{E,\text{max}}$):** To ensure the circuit can turn ON (emitter current can reach I_P at V_P):

$$R_{E,\text{max}} = \frac{V_{BB} - V_P}{I_P} = \frac{20\text{ V} - 13.7\text{ V}}{10 \times 10^{-6}\text{ A}} = \frac{6.3\text{ V}}{10\,\mu\text{A}} = \mathbf{630.0}\,\text{k}\,\Omega \tag{136}$$

- **Lower Bound ($R_{E,\text{min}}$):** To ensure the circuit can turn OFF (emitter current drops below I_V at V_V):

$$R_{E,\text{min}} = \frac{V_{BB} - V_V}{I_V} = \frac{20\text{ V} - 1.5\text{ V}}{8 \times 10^{-3}\text{ A}} = \frac{18.5\text{ V}}{8\text{ mA}} = \mathbf{2.31}\,\text{k}\,\Omega \tag{137}$$

Therefore, R_E must satisfy:

$$\mathbf{2.31}\,\text{k} < \mathbf{R_E} < \mathbf{630.0}\,\text{k}\,\Omega \tag{138}$$

The selected value $R_E = 33$ kΩ falls comfortably within this range.
Step 3: Calculate Timing Capacitor C_E for $f = 500$ Hz The period of oscillation is:

$$T = \frac{1}{f} = \frac{1}{500\text{ Hz}} = 0.002\text{ s} = 2.0\text{ ms} \tag{139}$$

The analytical period equation is:

$$T = R_E C_E \ln\left(\frac{V_{BB} - V_V}{V_{BB} - V_P}\right) \tag{140}$$

Evaluating the logarithmic argument:

$$\frac{V_{BB} - V_V}{V_{BB} - V_P} = \frac{20 - 1.5}{20 - 13.7} = \frac{18.5}{6.3} \approx 2.9365 \tag{141}$$

$$\ln(2.9365) = 1.0772 \tag{142}$$

Rearranging for C_E:

$$C_E = \frac{T}{R_E \times 1.0772} = \frac{2.0 \times 10^{-3}\text{ s}}{33 \times 10^3\,\Omega \times 1.0772} = \frac{2.0 \times 10^{-3}}{35548} = \mathbf{5.626 \times 10^{-8}}\text{ F} = \mathbf{56.26}\text{ nF} \tag{143}$$

Check with Standard Approximation: Using $T \approx R_E C_E \ln\left(\frac{1}{1-\eta}\right)$:

$$\ln\left(\frac{1}{1 - 0.65}\right) = \ln(2.857) = 1.0498 \implies C_E \approx \frac{2.0 \times 10^{-3}}{33000 \times 1.0498} = 57.73\text{ nF} \tag{144}$$

We select a standard 56 nF metallized polyester capacitor.

Problem 7: Power MOSFET Loss Breakdown and Thermal Sizing

Problem 7: High-Frequency MOSFET Loss Breakdown

An n-channel power MOSFET operates in a DC-DC buck converter stepping down a DC bus voltage of $V_d = 100$ V to power a server load. The switch conducts a flat-topped load current of $I_o = 15$ A with a duty cycle of $D = 0.40$ at a switching frequency of $f_{sw} = 100$ kHz.
From the device datasheet:

- On-state resistance at operating temperature: $R_{DS(on)} = 0.15\,\Omega$
- Turn-on current rise time: $t_r = 35$ ns

- Turn-off current fall time: $t_f = 50$ ns
- Output capacitance: $C_{oss} = 400$ pF
- Junction-to-case thermal resistance: $R_{\theta,jc} = 1.20°\text{C/W}$
- Case-to-sink thermal resistance: $R_{\theta,cs} = 0.30°\text{C/W}$
- Enclosure ambient temperature: $T_a = 40°\text{C}$; Maximum junction temperature: $T_{j,\max} = 125°\text{C}$

Determine:

(a) Conduction power loss (P_{cond}).

(b) Turn-on switching loss (P_{on}) and turn-off switching loss (P_{off}).

(c) Output capacitive discharge loss (P_{oss}).

(d) Total power dissipation (P_{total}).

(e) The maximum allowable heat sink thermal resistance ($R_{\theta,sa}$).

Solution:

Step 1: Calculate Conduction Loss (P_{cond}) In a buck converter with continuous inductor current, the switch current is a rectangular pulse of amplitude $I_o = 15$ A and duty cycle $D = 0.40$:

$$I_{D,rms} = I_o\sqrt{D} = 15 \times \sqrt{0.40} = 15 \times 0.63245 = 9.487 \text{ A} \tag{145}$$

$$P_{cond} = I_{D,rms}^2 \cdot R_{DS(on)} = (9.487 \text{ A})^2 \times 0.15\,\Omega = 90.0 \times 0.15 = \mathbf{13.50} \text{ W} \tag{146}$$

Step 2: Calculate Turn-On and Turn-Off Switching Losses Assuming linear transitions with clamped inductive switching:

$$P_{on} = \frac{1}{2}V_d \cdot I_o \cdot t_r \cdot f_{sw} = \frac{1}{2} \times 100 \text{ V} \times 15 \text{ A} \times (35 \times 10^{-9} \text{ s}) \times (100 \times 10^3 \text{ Hz}) \tag{147}$$

$$P_{on} = 750 \times 35 \times 10^{-9} \times 10^5 = 750 \times 3.5 \times 10^{-3} = \mathbf{2.625} \text{ W} \tag{148}$$

$$P_{off} = \frac{1}{2}V_d \cdot I_o \cdot t_f \cdot f_{sw} = \frac{1}{2} \times 100 \text{ V} \times 15 \text{ A} \times (50 \times 10^{-9} \text{ s}) \times (100 \times 10^3 \text{ Hz}) \tag{149}$$

$$P_{off} = 750 \times 50 \times 10^{-9} \times 10^5 = 750 \times 5.0 \times 10^{-3} = \mathbf{3.750} \text{ W} \tag{150}$$

Total switching loss:

$$P_{sw} = P_{on} + P_{off} = 2.625 + 3.750 = \mathbf{6.375} \text{ W} \tag{151}$$

Step 3: Calculate Capacitive Discharge Loss (P_{oss})

$$P_{oss} = \frac{1}{2}C_{oss} \cdot V_d^2 \cdot f_{sw} = \frac{1}{2} \times (400 \times 10^{-12} \text{ F}) \times (100 \text{ V})^2 \times (100 \times 10^3 \text{ Hz}) \tag{152}$$

$$P_{oss} = 200 \times 10^{-12} \times 10000 \times 100000 = 200 \times 10^{-12} \times 10^9 = \mathbf{0.20} \text{ W} \tag{153}$$

Step 4: Total Power Dissipation (P_{total})

$$P_{total} = P_{cond} + P_{sw} + P_{oss} = 13.50 + 6.375 + 0.20 = \mathbf{20.075} \text{ W} \tag{154}$$

Step 5: Maximum Permissible Heat Sink Thermal Resistance ($R_{\theta,sa}$)

$$R_{\theta,ja} = \frac{T_{j,\max} - T_a}{P_{total}} = \frac{125°\text{C} - 40°\text{C}}{20.075 \text{ W}} = \frac{85°\text{C}}{20.075 \text{ W}} = 4.234°\text{C/W} \tag{155}$$

$$R_{\theta,sa} \leq R_{\theta,ja} - (R_{\theta,jc} + R_{\theta,cs}) = 4.234 - (1.20 + 0.30) = 4.234 - 1.50 = \mathbf{2.734}°\text{C/W} \tag{156}$$

Problem 8: IGBT Switching Loss and Current Tail Analysis

Problem 8: IGBT Inverter Leg Losses and SiC Comparison

An Insulated Gate Bipolar Transistor (IGBT) operates in an Electric Vehicle (EV) 3-phase traction inverter leg supplied by a high-voltage DC link of $V_{dc} = 600$ V. During its on-time the transistor carries a flat-topped collector current of $I_{C,on} = 30$ A, with duty cycle $D = 0.50$, at a PWM switching frequency of $f_{sw} = 20$ kHz. (Its average current over the period is therefore $I_{C,avg} = D \cdot I_{C,on} = 15$ A.)
From the IGBT datasheet:

- On-state saturation voltage: $V_{CE(sat)} = 1.80$ V
- Turn-on energy loss per pulse: $E_{on} = 1.60$ mJ
- Turn-off energy loss per pulse: $E_{off} = 2.40$ mJ (which includes a prolonged recombination current tail of $t_{f2} = 350$ ns)

Determine:

(a) Conduction power loss (P_{cond}) and switching power loss (P_{sw}).

(b) Total power loss generated by the IGBT.

(c) A Silicon Carbide (SiC) MOSFET of equivalent 1200 V rating is evaluated as a replacement in the same leg (same current waveform, D and f_{sw}). It has $R_{DS(on)} = 40$ mΩ (at 100°C), $E_{on} = 0.45$ mJ and $E_{off} = 0.18$ mJ (no current tail). Calculate its RMS current, the new total power loss and the percentage reduction in dissipation.

Solution:
Step 1: IGBT conduction loss. The IGBT is modelled by its saturation drop, so the loss uses the *average* current (the duty cycle is already inside $I_{C,avg}$):

$$P_{cond,\text{IGBT}} = V_{CE(sat)} \cdot I_{C,avg} = 1.80\text{ V} \times (0.50 \times 30\text{ A}) = 1.80 \times 15 = \mathbf{27.0}\text{ W} \tag{157}$$

Step 2: IGBT switching loss. Total switching energy per cycle:

$$E_{sw} = E_{on} + E_{off} = 1.60 + 2.40 = 4.00\text{ mJ} \tag{158}$$

$$P_{sw,\text{IGBT}} = E_{sw} \cdot f_{sw} = (4.0 \times 10^{-3}\text{ J}) \times (20 \times 10^{3}\text{ Hz}) = \mathbf{80.0}\text{ W} \tag{159}$$

Step 3: Total IGBT dissipation.

$$P_{total,\text{IGBT}} = 27.0 + 80.0 = \mathbf{107.0}\text{ W} \tag{160}$$

Crucial observation: switching loss (80 W) is about 75% of the total heat, driven by the large turn-off tail energy ($E_{off} = 2.4$ mJ).
Step 4: SiC MOSFET. A MOSFET has no saturation-voltage knee, so its conduction loss is $I_{D,rms}^2 R_{DS(on)}$ with the RMS of the flat-topped current:

$$I_{D,rms} = I_{C,on}\sqrt{D} = 30 \times \sqrt{0.5} = 21.2\text{ A} \tag{161}$$

$$P_{cond,\text{SiC}} = (21.2)^2 \times 0.040 = 450 \times 0.040 = 18.0\text{ W} \tag{162}$$

$$P_{sw,\text{SiC}} = (0.45 + 0.18)\text{ mJ} \times 20\text{ kHz} = 0.63 \times 10^{-3} \times 20 \times 10^{3} = 12.6\text{ W} \tag{163}$$

$$P_{total,\text{SiC}} = 18.0 + 12.6 = \mathbf{30.6}\text{ W} \tag{164}$$

Step 5: Percentage reduction.

$$\frac{107.0 - 30.6}{107.0} \times 100\% = \mathbf{71.4}\% \tag{165}$$

This is consistent with the $60\% - 75\%$ inverter-loss reduction quoted for SiC in EV traction drives.
Note on high frequency. At $f_{sw} = 100$ kHz (to shrink motor ripple):

- IGBT: $P_{sw} = 4.0$ mJ $\times$ 100 kHz $= 400$ W, total ≈ 427 W (thermal burnout).
- SiC MOSFET: $P_{sw} = 0.63$ mJ $\times$ 100 kHz $= 63$ W, total ≈ 81 W.

Problem 9: High-Side Bootstrap Gate Driver Sizing

Problem 9: Bootstrap Driver Sizing for Half-Bridge Inverter

An IR2110 high-voltage gate driver IC drives an N-channel power MOSFET in the high-side position of a half-bridge converter. The gate driver supply rail is $V_{CC} = 15.0$ V.
Datasheet and circuit operating parameters:

- Total gate charge: $Q_g = 140$ nC
- External series gate resistance: $R_G = 6.8\,\Omega$; internal gate resistance: $R_{\text{int}} = 1.2\,\Omega$
- Bootstrap diode forward voltage drop: $V_{D,\text{boot}} = 0.70$ V
- Maximum high-side conduction time: $t_{on,\text{max}} = 30\,\mu$s (operating at minimum frequency $f_{\text{min}} = 25$ kHz)
- Driver quiescent current in high-side floating state: $I_{QBS} = 220\,\mu$A
- Level-shift charge per cycle: $Q_{LS} = 5$ nC
- Gate-source leakage current: $I_{GS,\text{leak}} = 100$ nA
- Maximum permissible voltage droop across the bootstrap capacitor during $t_{on,\text{max}}$: $\Delta V_{\text{boot}} = 0.50$ V

Calculate:

(a) The peak gate charging current ($I_{G,\text{peak}}$) at the onset of turn-on.

(b) The total electrical charge (Q_{total}) that the bootstrap capacitor must supply during one switching period.

(c) The absolute minimum bootstrap capacitance ($C_{\text{boot,min}}$) required to maintain gate drive stability.

Solution:
Step 1: Calculate Peak Gate Charging Current At the beginning of turn-on, the gate capacitance is fully discharged ($v_{GS} = 0$). The voltage across the bootstrap capacitor is approximately $V_{BS} = V_{CC} - V_{D,\text{boot}} = 15.0 - 0.70 = 14.30$ V. The total series loop resistance is $R_{total} = R_G + R_{\text{int}} = 6.8 + 1.2 = 8.0\,\Omega$.

$$I_{G,\text{peak}} = \frac{V_{BS}}{R_G + R_{\text{int}}} = \frac{14.30\text{ V}}{8.0\,\Omega} = \mathbf{1.7875}\text{ A} \approx \mathbf{1.79}\text{ A} \tag{166}$$

Takeaway: Even though average gate power is milliwatts, the gate driver must be rated to source an instantaneous peak current of nearly 1.8 **A**!
Step 2: Calculate Total Charge Supplied per Cycle (Q_{total}) The bootstrap capacitor must deliver charge to four distinct sinks during the high-side on-state:

1. Gate charge required to fully charge C_{iss}: $Q_g = 140$ nC.

2. Driver IC floating-section quiescent consumption:

$$Q_{QBS} = I_{QBS} \cdot t_{on,\text{max}} = (220 \times 10^{-6}\ \text{A}) \times (30 \times 10^{-6}\ \text{s}) = 6.60 \times 10^{-9}\ \text{C} = 6.60\ \text{nC} \quad (167)$$

3. Internal level-shift circuit charge: $Q_{LS} = 5.0$ nC.

4. Gate-source and diode reverse leakage:

$$Q_{leak} = I_{GS,\text{leak}} \cdot t_{on,\text{max}} = (100 \times 10^{-9}\ \text{A}) \times (30 \times 10^{-6}\ \text{s}) = 0.003\ \text{nC} \approx 0 \quad (168)$$

Summing all charge components:

$$Q_{\text{total}} = Q_g + Q_{QBS} + Q_{LS} + Q_{leak} = 140 + 6.60 + 5.0 + 0.003 = \mathbf{151.603}\ \text{nC} \quad (169)$$

Step 3: Calculate Minimum Bootstrap Capacitance ($C_{\text{boot,min}}$) From the fundamental capacitance definition $Q = C\Delta V$:

$$C_{\text{boot,min}} = \frac{Q_{\text{total}}}{\Delta V_{\text{boot}}} = \frac{151.603 \times 10^{-9}\ \text{C}}{0.50\ \text{V}} = 3.032 \times 10^{-7}\ \text{F} = \mathbf{0.3032}\,\mu\text{F} \quad (170)$$

Applying a conservative engineering margin of $2\times$ to account for ceramic capacitor DC-bias capacitance derating and temperature variation:

$$C_{\text{boot, recommended}} \geq 2 \times 0.3032\,\mu\text{F} \approx \mathbf{0.68}\,\mu\text{F} - \mathbf{1.0}\,\mu\text{F} \quad (171)$$

A $1.0\,\mu$F, 25 V X7R low-ESR ceramic capacitor placed directly adjacent to the V_B and V_S pins will guarantee optimal performance.

Master Comparison Tables & Quick-Reference Formulas

Comprehensive Power Semiconductor Device Matrix

Master Quick-Reference Formula Sheet

Master Formula Collection for Examination Preparation

1. **Thermal Modeling & Heat Sink Sizing:**
   - Steady-state thermal Ohm's law: $T_j - T_a = P_D \cdot R_{\theta,ja} = P_D \cdot (R_{\theta,jc} + R_{\theta,cs} + R_{\theta,sa})$
   - Heat sink sizing: $R_{\theta,sa} \leq \frac{T_{j,\text{max}} - T_a}{P_D} - (R_{\theta,jc} + R_{\theta,cs})$
   - Operating case temperature: $T_c = T_j - P_D \cdot R_{\theta,jc} = T_a + P_D \cdot (R_{\theta,cs} + R_{\theta,sa})$
   - Operating sink temperature: $T_s = T_c - P_D \cdot R_{\theta,cs} = T_a + P_D \cdot R_{\theta,sa}$
   - Maximum power without heat sink: $P_{D,\text{max(no sink)}} = \frac{T_{j,\text{max}} - T_a}{R_{\theta,jc} + R_{\theta,ca}}$
2. **Power Losses in Semiconductor Switches:**
   - General Conduction Loss: $P_{cond} = V_0 \cdot I_{avg} + r_d \cdot I_{rms}^2$
   - MOSFET Conduction Loss: $P_{cond,\text{MOSFET}} = I_{D,rms}^2 \cdot R_{DS(on)}$
   - IGBT Conduction Loss: $P_{cond,\text{IGBT}} \approx V_{CE(sat)} \cdot I_{C,avg}$ ($I_{C,avg} = D\,I_{C,on}$ for a flat-top current; do not multiply by D again)
   - Clamped-inductive switching loss: $P_{sw} = \frac{1}{2} V_d I_L \left(t_{c(on)} + t_{c(off)}\right) f_{sw} = (E_{on} + E_{off}) f_{sw}$ (crossover intervals $t_{c(on)} = t_{ri} + t_{fv}$, $t_{c(off)} = t_{rv} + t_{fi}$; exam shortcut: use datasheet t_r, t_f). Resistive load: $\frac{1}{6} V_d I_L t\, f_{sw}$ per edge

Table 7: The Master Power Semiconductor Device Comparison Matrix

Parameter	Power Diode	Thyristor (SCR)	Power BJT	Power MOSFET	IGBT	SiC MOSFET	GaN HEMT
Controllability	Uncontrolled	Semi-controlled (ON only)	Fully controlled	Fully controlled	Fully controlled	Fully controlled	Fully controlled
Carrier Type	Bipolar (Minority)	Bipolar (Minority)	Bipolar (Minority)	Unipolar (Majority)	Bipolar (Minority)	Unipolar (Majority)	Unipolar (2DEG)
Drive Type	None	Gate current pulse	Continuous base current	Voltage controlled (V_{GS})	Voltage controlled (V_{GE})	Voltage controlled (V_{GS})	Voltage controlled (V_{GS})
Drive Power	Zero	Low (pulsed)	High ($h_{FE} \sim 5-10$)	Very low (dynamic only)	Very low (dynamic only)	Very low (dynamic only)	Ultra-low (Q_g tiny)
Max Voltage	Up to 5000 V	Up to 8000 V	Up to 1400 V	Up to 1000 V	Up to 6500 V	Up to 3300 V+	Up to 650 V − 900 V
Max Current	Up to 5000 A	Up to 5000 A	Up to 400 A	Up to 200 A	Up to 3000 A	Up to 300 A	Up to 150 A
Forward Drop	$0.9-1.5$ V	$1.2-2.0$ V	$1.0-2.0$ V	$I_D \cdot R_{DS(on)}$ (High at >200V)	$1.5-2.5$ V (Low)	$I_D \cdot R_{DS(on)}$ (Very low)	$I_D \cdot R_{DS(on)}$ (Ultra-low)
Temp. Coeff.	Negative	Negative	Negative	Positive (Self-balancing)	Positive at rated I_C	Positive	Positive
Switching Freq.	$\leq$ 1 kHz (Line) / 50 kHz (FRD)	$\leq$ 1 kHz (Line)	≤ 10 kHz	Up to 1 MHz	$\leq 20-30$ kHz (Tail limit)	100 kHz − 500 kHz	500 kHz − 10 MHz
Turn-Off Method	Natural Commutation	Line / Forced Commutation	Base current removal	Gate discharge to 0V	Gate discharge to 0V	Gate discharge to -4V	Gate discharge to 0V
Second Breakdown	No	No	Yes (Severe hazard)	No	No	No	No
Paralleling	Difficult (Needs sharing chokes)	Difficult (Needs sharing resistors)	Difficult (Thermal runaway)	Easy (Self-balancing)	Easy (for modern NPT/FS)	Easy	Easy
Reverse Recovery	Yes (Q_{RR} significant)	Yes ($t_{rr}+t_{gr}$)	Storage delay time	Yes (Internal body diode)	None (Needs antiparallel diode)	Very small	Zero ($Q_{rr}=0$)
Primary Domain	Mains rectifiers, welding	HVDC, heavy phase control, soft starters	Obsolete (Replaced by IGBT/MOSFET)	Low voltage (<200V), SMPS, Automotive	Medium/High power (>600V), VFDs, EV	EV traction, solar inverters, >800V	Fast chargers, servers, RF, telecom

- Output Capacitive Loss: $P_{oss} = \frac{1}{2} C_{oss} V_d^2 f_{sw}$

3. **Power Diode Reverse Recovery:**
   - Reverse recovery time: $t_{rr} = t_a + t_b = (1+S)\, t_a = \sqrt{\frac{2Q_{RR}(1+S)}{di/dt}}$ $\quad (\approx \sqrt{2Q_{RR}/(di/dt)}$ only for $S \to 0)$
   - Peak reverse current: $I_{RR} = \left(\frac{di}{dt}\right) t_a = \sqrt{2Q_{RR}\left(\frac{di}{dt}\right)\left(\frac{1}{1+S}\right)}$
   - Softness factor: $S = t_b/t_a$
   - Stored charge: $Q_{RR} = \frac{1}{2} I_{RR} t_{rr}$
   - Reverse recovery power loss: $P_{rr} \approx \frac{1}{6} V_R I_{RR} t_b f_{sw}$
   - Peak inductive turn-off spike: $v_{\text{spike}} = V_R + L_\sigma \left|\frac{di}{dt}\right|_{t_b}$
4. **SCR Physics & Commutation:**
   - Two-transistor anode current: $I_A = \frac{\alpha_2 I_G + I_{CBO1} + I_{CBO2}}{1-(\alpha_1+\alpha_2)}$
   - Latching condition: $\alpha_1 + \alpha_2 \longrightarrow 1.0$
   - Device turn-off time: $t_q = t_{rr} + t_{gr}$
   - Golden commutation condition: $t_c > t_q$ $\quad$ (typically $t_c \geq 1.5 - 2.0 \times t_q$)
   - Minimum gate pulse width with $R-L$ load: $t_{p,\text{min}} \approx \frac{L \cdot I_L}{V_s}$
5. **SCR Protection & Snubber Design:**
   - di/dt series inductor: $L_s \geq \frac{V_m}{(di/dt)_{\text{max}}}$ (and $L_s \geq \frac{V_m^2}{I_{T,\text{max}}(dv/dt)_{\text{max}}}$)

- Initial dv/dt across SCR: $\frac{V_m R_s}{L_s} \leq (dv/dt)_{\max}$, i.e. $R_s \leq \frac{L_s (dv/dt)_{\max}}{V_m}$
- Discharge current limit on R_s: $R_s \geq \frac{V_m}{I_{T,\max}}$
- Snubber capacitor for damping $\xi \approx 0.65$: $C_s = L_s \left(\frac{2\xi}{R_s}\right)^2$ (equivalently $R_s = 2\xi\sqrt{L_s/C_s}$)
- Snubber resistor power dissipation: $P_{\text{snub}} = C_s \cdot V_m^2 \cdot f$

6. **UJT Relaxation Oscillator:**
   - Intrinsic standoff ratio: $\eta = \frac{R_{B1}}{R_{B1}+R_{B2}} = \frac{R_{B1}}{R_{BB}}$
   - Peak point voltage: $V_P = \eta V_{BB} + V_D$
   - Timing period: $T = R_E C_E \ln\left(\frac{V_{BB}-V_V}{V_{BB}-V_P}\right) \approx R_E C_E \ln\left(\frac{1}{1-\eta}\right)$
   - Firing frequency: $f = \frac{1}{T} \approx \frac{1}{R_E C_E \ln\left(\frac{1}{1-\eta}\right)}$
   - Resistance bounds for oscillation: $\frac{V_{BB}-V_P}{I_P} > R_E > \frac{V_{BB}-V_V}{I_V}$
7. **Gate Drivers & Wide Bandgap FOM:**
   - Peak gate sourcing current: $I_{G,\text{peak}} = \frac{V_{CC}-V_D}{R_G+R_{\text{int}}}$
   - Minimum bootstrap capacitance: $C_{\text{boot}} \geq \frac{Q_g + I_{QBS} \cdot t_{on,\max} + Q_{LS}}{\Delta V_{\text{boot}}}$
   - Baliga's Figure of Merit (BFOM): BFOM $= \epsilon_s \cdot \mu_n \cdot E_c^3$
   - Specific on-resistance (ideal, constant E_c): $R_{on,sp} = \frac{4V_{BR}^2}{\epsilon_s \mu_n E_c^3}$

END OF MODULE 1 MASTER STUDY GUIDE

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